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DIGITAL TRANSFORMATION IN ORGANIZATIONS: A MULTIPAPER INVESTIGATION IN THE BRAZILIAN RETAIL INDUSTRY

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DIGITAL TRANSFORMATION IN ORGANIZATIONS: A MULTIPAPER INVESTIGATION IN THE BRAZILIAN RETAIL INDUSTRY

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FLAVIA DE PAIVA MICHELOTTO

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I dedicate this dissertation to my mother, without whom I wouldn't have made it this far, and to my children, without whom I wouldn't have had reasons to make it this far. I love you deeply.

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ABSTRACT

Digital Transformation (DT) encompasses a range of changes in society and modern organizations, driven by innovations accrued from the adoption of emerging Information and Communication Technologies (ICT). Indeed, DT has enabled fundamental shifts in business strategies and the integration of ICT with organizational structures, thereby attracting significant interest in academia. Moreover, given its rapid and ongoing evolution, the concept of DT requires continuous monitoring and in-depth examination. In fact, DT's multifaceted characteristic has contributed to the absence of a unified definition for it, particularly within the business literature. In this sense, contributing to this evolution and understanding of how DT is perceived and assessed in organizations in the context of an emerging country, this doctoral dissertation seeks to deepen the discussion on the theme through a multipaper investigation, proposing three unexplored approaches that are linked under the umbrella-concept of Organizational Digital Transformation (ODT). The first paper is a literature review of Digital Transformation readiness in organizations, aiming to conceptualize and investigate how to assess organizations' readiness for DT initiatives. As a result, the paper proposes an integrated model for Organizational Digital Transformation Readiness (ODTR) that encompasses not only technological factors but also cultural and human aspects, among others, highlighting the multidisciplinary nature of DT. The second paper is a subsequent step, drawing on the main antecedents for ODTR outlined in the literature review. The research aims to investigate the critical success factors that enable DT in organizations using a Delphi-based approach that ranks the degree of importance of these factors and identifies new ones. This study was performed with retailers' experts in Brazil, an emerging country with considerable socioeconomic disparities. The retail sector was chosen because of its substantial socioeconomic influence and the significant recent transformations it has undergone. The results reveal that most critical success factors are related to management capability and business processes rather than to technology autonomously. Following the theme, the third paper investigates another aspect of ODT, by investigating how retailers perceive and deal with recent innovations in the payment model in Brazil. That way, the Social Representation Theory (SRT) is used to understand the new digital peer-to-peer payment system, Pix, and why it has become a primary successful payment tool in the retail sector. Thus, the three papers aim to contribute to the academic debate on DT from a business perspective, provide unexplored managerial outcomes for retailers, and guide other practitioners in their future DT efforts.

Keywords: Digital Transformation; Organizational Digital Transformation Readiness; Digital Payment; Mobile Payment; Retail; Pix.

LIST OF ABBREVIATIONS

ABRAS	Associação Brasileira de Supermercados *(Brazilian Supermarket Association)*
AI	Artificial Intelligence
BI	Business Intelligence
CMMI	Capability Maturity Model Integration
CNDL	Confederação Nacional de Dirigentes Lojistas *(National Confederation of Retailers)*
CX	Customer Experience
DREAMY	Digital Readiness Assessment Maturity model
DT	Digital Transformation
ICT	Information and Communication Technology
IoT	Internet of Things
IS	Information Systems
KPI	Key Performance Indicators
NRF	National Retail Federation
ODT	Organizational Digital Transformation
ODTR	Organizational Digital Transformation Readiness
PCM	Perceived Critical Mass
SME	Small and Medium Enterprises
SR	Social Representations
SRT	Social Representation Theory
TAM	Technology Acceptance Model
UTAUT	Unified Theory of Acceptance and Use of Technology

PREFACE

This dissertation has been submitted to fulfill the requirements of the Doctoral Program in Administration at Getulio Vargas Foundation (FGV EBAPE), totally held in English. It includes an introduction, one submitted paper (paper I), one conference paper (paper II), one paper to be submitted (paper III), and a conclusion.

Paper I (under review at the *Industrial Management & Data Systems* journal)

Michelotto, F.; Joia, L. A. (2024). Organizational Digital Transformation Readiness: A Literature Review.

Paper II (accepted at the *XLVIII Encontro da ANPAD - EnANPAD 2024* conference)

Michelotto, F.; Joia, L. A. (2024). Critical Success Factors for Organizational Digital Transformation: A Delphi Study in the Retail Industry.

Paper III (to be submitted)

Michelotto, F.; Joia, L. A. (2024). Digital Transformation in Payment Methods: The Social Representation of Pix in the Retail Industry.

Additionally, different pieces of work were carried out during the doctoral course. Even if not entirely linked to the dissertation topic, they contributed to the overall academic development. The main works, including reviewing papers, culminated in the following publications in journals and annals of events:

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Michelotto, F., & Joia, L. A. (2023). Unveiling the smart city concept: Perspectives from an emerging market via the Social Representation Theory. *Sustainability*, 15(10), 8155.

Joia, L. A., Michelotto, F., & Lorenzo, M. (2022). Sustainability and the Social Representation of the COVID-19 Pandemic: A Missing Link. *Sustainability*, 14 (17), 10527.

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Michelotto, F., & Joia, L. A. (2021). The Social Representation of Smart Cities: A View from Brazil. In *Electronic Government: 20th IFIP WG 8.5 International Conference, EGOV 2021, Granada, Spain, September 7–9, 2021, Proceedings 20* (pp. 217-228). Springer International Publishing.

Joia, L. A., & Michelotto, F. (2020). Universalists or utilitarianists? The social representation of COVID-19 pandemic in Brazil. *Sustainability*, 12(24), 10434.

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1. Introduction

The current investigation is a comprehensive dissertation project comprising a set of three papers developed over the doctoral program. The papers comprising this collection share a consistent theme and employ sundry theoretical approaches that complement each other, allowing a fluid study of prominent discussions over the organizational DT theme. Besides, the dissertation uses qualitative-quantitative methodologies and comparable empirical settings to explore how organizations prepare themselves for, perceive, and assess DT initiatives.

Over the last decades, there has been a significant move toward the digitalization of business infrastructure. Besides, DT is closely linked to organizations, as it focuses on the impact of ICT on various aspects of the organizational structure, routines, information flow, and capabilities (Vial, 2019; Li, 2020). Thus, DT has become a top priority for many organizations on their leadership agenda (Matt et al., 2015), as they seek to remain competitive in a rapidly changing technological landscape. This revolution has had a profound impact on many organizations, as it has transformed processes and business models in almost all industries through the implementation of digital technologies to create value (Demirkan et al., 2016; Lanzolla et al., 2020). As a result, there has been a fundamental shift in business strategies, methods, and capabilities, from the products and services offered to the rise of Artificial Intelligence (AI) tools that started being implemented in companies (Bharadwaj et al., 2013; Kitsios & Kamariotou, 2021). However, despite the growth of DT research, several aspects remain underdeveloped, being not yet understood how business strategy relationships lead to successful implementation of DT initiatives in organizations (Lanzolla et al., 2020; Mishra et al., 2023).

Indeed, companies aiming to implement DT initiatives may unveil critical questions about how to get started and which barriers can hamper the success of DT endeavors. These questions directly relate to the company's readiness to be digitally transformed, focusing on its internal capabilities and barriers (Stoianova et al., 2020). In fact, digital readiness represents a specific scenario of change in which digital technologies play a crucial role in operationalizing - through technology, attitudes, and capabilities – successful DT initiatives (Moeini Gharagozloo et al., 2021; Gfrerer et al., 2021; Pirola et al., 2018).

A subsequent level to assess the success of DT endeavors in organizations is related to enlisting their digital capabilities (Henriette et al., 2016) and their critical factors. To achieve this objective, companies' leaders must redefine their roles and responsibilities aligned with the DT process, defining straightforward strategies that consider digital skills and capabilities to achieve new objectives (Hai et

al., 2021). Studying DT critical success factors is pivotal to understanding how companies achieve competitive advantages and further prosperity (Dinter, 2013).

In order to address the current gaps in the scientific literature, this study explores the progression of DT processes within organizations. It conducts a detailed examination of the preparatory steps for DT readiness, the critical success factors and their relevance for DT, and, finally, explores a practical application of DT by understanding the social representation (SR) of a new digital payment method and how companies perceive and adopt it. For this endeavor, the retail sector was chosen due to its profound socioeconomic impact and the unique transformations it has undergone in recent years, particularly in the digital realm. This highlights the sector's relevance for our study.

Finally, Brazil, the research ground, is a blend of two worlds: highly developed and very poor societies coexisting. This unique duality is mirrored in the retail/commerce businesses, ranging from high-end stores and e-commerce platforms to modest street commerce. Besides, in the 1990s, Brazil underwent significant economic reforms, liberalizing markets and reducing wage inequality, unlike other Latin American countries, thanks to the inflow of more educated workers (Firpo & Portella, 2019). Thus, these diverse characteristics of the Brazilian retail segment point to a valuable, diverse and experienced backdrop for this present study.

1.1 Organizational Digital Transformation

The fast proliferation of digital technologies has given rise to the complex concept of DT. This construct has taken on a multitude of approaches in the literature, with numerous definitions, and continues to be a subject of intense debate in both academic and practitioner circles (Gong & Ribiere, 2021). DT is strongly related to organizations, addressing and underscoring the influence of IT on organizational structure, routines, information flow, and capabilities (Vial, 2019; Li et al., 2017). It is, therefore, a dynamic process that encompasses the interconnections between and within organizations, necessitating their readiness to implement action plans based on technological possibilities (Westerman, 2017).

Becoming digital is the most omnipresent challenge companies have faced in the last few decades. However, this transformation process is sustained by more than technology. Previous studies have identified that successful DT initiatives are associated not only with the use of technology. Orlikowski (1996) found empirical evidence that organizational transformation was enabled by technology but not caused by it. Researchers and organizations gradually perceived that digital transformation was not only a matter of technology alone (Henriette et al., 2015). Non-technological factors such as culture, leadership, and training are pivotal in enabling a successful transformation.

Both academics and practitioners acknowledge the importance of human resources in this process (Gong & Ribiere, 2021). The lack of digital skills is often cited as a limitation to the success of DT (Agrawal et al., 2019). Therefore, the key to successful DT initiatives lies in prioritizing the transformational dimension over the digital aspects (Lam & Law, 2019). This approach combines the use of technology with skilled workers to fully leverage the transformative potential of digital possibilities (Nadkarni & Prugl, 2020).

Developing theories around DT is a challenging task for researchers, just as practitioners face difficulties asserting their authority in implementing DT strategies in organizations (Gong & Ribiere, 2021). Different maturity models for DT were developed prioritizing industrial aspects, however, there remains a gap in readiness considering the business' aspects (Aras & Büyüközkan, 2023). Moreover, there are also gaps in understanding how business strategies lead to successful DT initiatives in organizations (Mishra et al., 2023).

1.2 Retail Industry in a Developing Country, Brazil

Due to the fast advancements in technology impacting the retail industry, experts and professionals in retailing aim to comprehensively understand the scope and nature of these effects (Shankar et al., 2021). In the 1990s, Brazil, an emerging economy with significant socioeconomic disparities, experienced a major transformation in its retail sector. Traditionally dominated by small family-run stores, the sector began allowing foreign direct investment, potentially setting the stage for the introduction and growth of retail chains (Reardon et al., 2003; Vries, 2008).

The Brazilian retail industry has been shaped by various technological advancements, especially trying to promote digital inclusion, and increasing online sales channels, which intensified competition among retail firms (Sousa et al., 2021). The most prominent Brazilian retail companies have implemented sophisticated information and management technologies to maintain their market share and extend their reach (Sousa et al., 2021). Brazil is the only country in Latin America that ranks among the ten largest e-commerce markets in the world (Botelho & Guissoni, 2016).

Among different retail segments, the Supermarket/Grocery leads the percentage of consumer sales in Brazil, representing more than 36% of the consumption, followed by the Automotive segment, with 27%, Building Supply Stores/DYI, with 8.6%, and Apparel, with 7.5% (IBEVAR, 2022). The French-based Carrefour Group leads the companies' ranking, responding to annual revenue of US$15.3 billion in 2022 (IBEVAR, 2022) in a highly concentrated market.

The Supermarket/Grocery segment in Brazil contributes to 7% of the national GDP (of US$1.6 trillion) and provides income to 3.1 million individuals who directly or indirectly participate in an

ecosystem that caters to 28 million consumers per day, according to the 2022 ranking from the Brazilian Supermarket Association (ABRAS) (ABRAS, 2023; World Bank Group, 2023). However, while larger retail corporations have started to increase their usage of e-commerce, social media marketing or analytics-based supply-chain management, many small and regional companies have not adequately developed capabilities to fulfill the present demands (Begley et al., 2020).

The rise of e-commerce in Brazil started in 1996, and one of the recent factors contributing to its evolution has been the emergence of Pix for online purchases. Pix is the new peer-to-peer digital payment tool the Brazilian Central Bank launched in November 2020 (Lobo & Brandt, 2021). In early 2021, only 1% of online purchases were made using Pix, but by 2023, this had increased to 26.2% (source: IstoéDinheiro, 2024). Across all retail transactions, including physical stores, Pix accounted for approximately 35% of payments in the first quarter of 2023 (Supervarejo, 2023), signifying its significant role in retail payments. Notwithstanding, Pix is recognized as a worldwide model of financial inclusion (Mercado e Consumo, 2024).

2. Research Question and Objectives

Throughout the growing academic literature on DT in the organizational field, most specifically in the retail industry, there are remaining gaps to be addressed. Therefore, by adopting the premise that DT is a continuous process, this dissertation aims to understand how organizations navigate the DT journey, from preparation and readiness for change through defining critical success factors to understanding how a DT process is represented and, moreover, assimilated in a business environment. Accordingly, the research question of this dissertation aims to consolidate these approaches by investigating *"How Digital Transformation has advanced within organizational environments, from evaluating their attributes for readiness to understanding how they manage and assess critical success factors in actual DT processes?"*

To find answers to the proposed research question and attempt to contribute to the extant literature, the following objectives arise:

a) Structure and propose an integrative framework for assessing an organization's readiness for successful digital transformation by reviewing existing literature on digital readiness.

b) Analyze the critical success factors of DT in organizations, ranking them in order of importance using the Delphi method.

c) Analyze how an actual DT process in retail organizations is perceived by exploring the details of the new digital payment method, Pix, using the Social Representation Theory.

3. Methodological Procedures

The investigation employed a mixed methods approach in the three papers, utilizing both qualitative and quantitative methods to explore how organizations prepare themselves for and assess DT initiatives.

Paper I encompasses a literature review that investigates the constructs that comprise organizational digital transformation readiness. The search aimed to cover articles published in journals associated with the theme in the Web of Science database, as well as in the most important Information Systems (IS) journals from 2004 to 2021 – considering that the literature is restricted before this period. Albeit there was a search for the total number of articles on the "digital transformation" construct, as the number of articles is much broader, the analyses were performed considering the following three focal constructs "e-readiness"; "digital readiness"; and "digital transformation readiness". The subsequent analysis considered the areas of interest in which they were published, allowing the identification of the main categories per theme, which helped to prioritize the themes that are related to IS, ICT, and other organizational themes. The relevant themes related to DT readiness were then used to build a concept matrix to compare the findings in the literature. Finally, a framework was developed to assess the organizational readiness for DT, by investigating the logical order and connections of categorized themes selected in the reviewed articles.

Paper II uses the Delphi method to investigate the main drivers for successful DT in organizations. The Delphi approach was employed considering the absence of direct communication among the respondents via an online survey formulated on the Qualtrics platform, by which retailer executives and experts were presented to twelve statements related to the five dimensions accrued from the ODTR literature review, aiming to assess their level of agreement on the critical factors previously outlined. They were asked to evaluate, rank in order of significance and suggest other possible critical success factors associated with DT.

Paper III employed both qualitative and quantitative methods to determine the social representation and overall perceptions of retailers regarding the new peer-to-peer instant mobile payment tool Pix. The first step consisted of conducting a survey divided into three parts: (1) free words evoked and the reasons for the chosen words, (2) open questions about the understanding of Pix in the organizational process of DT and (3) socio-demographic data related to the participants. Finally, to assemble the social representation, the participants' evocations were listed to conduct a lexicographical analysis (frequency of occurrence of terms) and the analysis of the order of appearance of the terms evoked in order to identify the Central Nucleus and the Peripheral System of the social

representation. Two subsequent analyses were then performed to confirm the findings depicted in the Vergès' quadrant, namely similarity analysis - used to outline social representations by analyzing the frequency and the order of evocation via implicative statistics - and content analysis - with the open responses detailing the participants' reasons for choosing the words evoked. That done, a textual corpus was organized to construct a word cloud - a technique that displays a diagram that correlates the font size of the tags with the word frequencies.

4. Dissertation structure

This multipaper dissertation comprises three papers and conclusions to answer to the research question and to address the objectives proposed in section 2: structuring an integrative framework related to the organizational readiness for a successful DT; analyzing how the critical success factors for organizational DT are ranked in order of importance and assessed; and analyzing how an actual DT process in retail organizations is perceived by exploring an actual DT process brought by the new digital payment method, Pix. The work was developed during the doctoral program from July 2020 to June 2024.

Paper I provides a systematic literature review and a framework for Organizational Digital Transformation Readiness (ODTR), which is conducted by investigating the pivotal attributes that allow companies to be prepared to achieve DT. Paper II investigates the critical success factors that enable DT in organizations by ranking their degree of importance via a Delphi-based approach. Paper III identifies the social representation of the digital peer-to-peer payment method Pix, according to the lenses of the Social Representation Theory.

The structure of the work is detailed in Figure 1, and the papers composing it follow in sequence.

Digital Transformation in Organizations: a Multi-Paper Investigation in the Brazilian Retail Industry			
Research Question	How Digital Transformation has advanced within organizational environments, from evaluating their attributes for readiness to understanding how they manage and assess critical success factors in actual DT processes?		
Papers	**Paper I** Organizational Digital Transformation Readiness Literature Review	**Paper II** Critical Success Factors for Organizational Digital Transformation: A Delphi Study in the Retail Industry	**Paper III** Digital Transformation in Payment Methods: The Social Representation of Pix in the Retail Industry
Objectives	Identify in the literature the key constructs for organizational DT readiness.	Understand how organizations rank and assess the critical success factors for DT.	Understand how organizations perceive and conceptualize a real process of DT- the new digital payment model- Pix.
Methods	Literature Review	Delphi Method	Social Representation Theory
Subjects	Existing Literature	Retail Industry and Experts	Retail Industry and Experts
Outcomes	ODTR Framework Proposition	Digital Transformation Critical Success Factors' Ranking	Social Representation of Digital Payment Pix

Figure 1. Dissertation Structure.

5. Paper I

ORGANIZATIONAL DIGITAL TRANSFORMATION READINESS: A LITERATURE REVIEW

Abstract

The widespread adoption of digital technologies across various aspects of society has led to the emergence of two highly context-dependent constructs that explain how businesses are getting ready for transformative digital processes: digital readiness and digital transformation (DT). However, exploring how organizational readiness influences DT lacks a clear definition. This research addresses this gap by working towards a practical definition and model of Organizational Digital Transformation Readiness (ODTR). The authors investigate how digital transformation readiness in organizations ought to be developed and conceptualized through a systematic literature review. The literature review shows the similarity between e-readiness and digital readiness, indicating an organization's maturity for transformation. Organizational readiness encompasses diverse factors not only technological, including culture, human aspects, business models, and sustainability, underscoring its multidisciplinary nature. This literature review synthetizes the literature on digital transformation readiness for organizations and proposes an integrative framework for assessing the ODTR. It provides synthesized insights and practical recommendations.

Keywords: Digital Transformation, Digital Transformation Readiness, Organizational Digital Transformation Readiness, Systematic Literature Review.

1. Introduction

Information and communication technology (ICT) has been a fundamental structural part of modern society (Dobrota et al., 2012), playing a significant role in promoting socio-economic development (Brynjolfsson & McAfee, 2011; Milenkovic et al., 2016). In fact, in the recent decades, significant advancements in ICT have made available several functionalities that allowed firms to adapt their business infrastructure to an entirely new digital era (Bharadwaj et al., 2013). Due to that, a new concept has just emerged in academia and among practitioners, namely Digital Transformation (DT) (Wessel et al., 2021). DT is a social construction, implying in a cultural transformation since it represents a process of social interaction and collective learning by which people embrace and disseminate digital technologies (Švarc et al., 2020). The notion of DT has become ubiquitous and is often understood as a change in how a company employs digital technologies to develop new digital business models that help same generating more value (Lanzolla et al., 2020). Indeed, the digital revolution has transformed the nature of many organizations by radically changing processes and business models in almost all industries (Demirkan et al., 2016). In this way, organizations have made an effort to respond to the challenges of competition and globalization brought about by the rapid and profound changes resulting from digital technologies (Reis et al., 2018), under penalty of, by not doing so, not being able to survive in the market (Nasution et al., 2018). Thus, DT has been an imperative for organizations to remain competitive in the current digital world (Vial, 2019).

However, to date, there is not a unique and commonly accepted definition for the construct "digital transformation", which is often viewed simplistically as an ongoing process of adopting emerging technologies for significant change (Teichert, 2019). Besides, although DT has emerged as a phenomenon in the Information Systems (IS) area in recent years (Vial, 2019), there is as yet few articles that jointly address the constructs needed to define organizational digital transformation readiness. However, understanding the steps and background needed to successfully implement DT in organizations is crucial in an increasingly ICT-dependent world. Thus, for DT initiatives to be successful in companies, they must be adequately ready for it (Gfrerer et al., 2021).

The term "readiness" refers to "the state of being ready or prepared for something" (Oxford, 2021, n.d.), describing a forward-looking state for something that is coming or has almost arrived (Voß & Pawlowski, 2019). E-readiness specifically represents the transformation of society to deal with complex issues related to the digital world (Alaaraj & Ibrahim, 2014). In fact, since its inception, the concept of e-readiness has encompassed a wide range of definitions and different measurement methods, involving sundry approaches and results (Alghamdi et al., 2011; Choucri et al., 2003; Molla et al., 2008; Sheoran & Vij, 2022). Successful organizational DT initiatives require companies to be ready to implement or develop different capabilities, according to the business context in which they operate (Reis et al., 2018).

Moreover, so far, some questions regarding DT in organizations have lacked further development, remaining understudied (Stief et al., 2016; Vial, 2019; Lanzolla et al., 2020), such as: *How do DT initiatives take place in organizations? What organizational requirements (resources, capabilities, processes) are needed and what is the sequence of actions to promote successful DT in organizations?* In line with this, the object of this study is to investigate: *What do companies need to succeed in DT initiatives, that is, how to assess the DT readiness of organizations?* Thus, to contribute to the extant literature on the subject, this article aims to review the topic of DT readiness and propose a framework for successful implementation of DT initiatives in organizations. From this perspective, a range of definitions, concepts and connections referring to what is understood by DT, with a focus on organizations, will be analyzed.

2. The Foundations of Digital Transformation

According to its most basic meaning, transformation refers to "a complete change in somebody/something" (Oxford, 2022, n.d.). In this sense, the DT construct would encompass a complete change by means of digital devices, technologies and/or contexts. In the early 1950s, in one of the first publications that mentioned DT, the expression was approached in a very basic way, far from the complexity that has been assigned to it recently. Davis (1954 p.90) proposed that "a great deal of progress has been made in converting analog data into a useful digital form for data processing. The way in which this analog to DT is accomplished is not biased by tradition or legal reasons." As society evolved, with the fast-growing use of digital technologies not only in machines or industries but in the most trivial aspects of daily life, the construct gained multifaceted approaches in the scientific literature, with several definitions that emerged as a buzzword and is affecting its credibility and causing a turmoil in both academic and practitioner fields (Gong & Ribiere, 2021).

The literature on DT is quite diverse, involving several knowledge areas, which has led the construct to not have a consolidated definition yet (Nadkarni & Prügl, 2021). The initial studies on DT were related to the strategic role that ICT used to play to support business strategies (Kochan & Useem, 1992). As DT has different meanings for different audiences, this concept has not been equably applied in the Information Systems literature, being so associated with different perspectives (Voß & Pawlowski, 2019). Probably one of the most comprehensive works on the concept of DT was carried out by Vial (2019, p. 118), who defined DT as "a process that aims to improve an entity by triggering significant changes in its properties through combinations of information, computing, communication, and connectivity technologies." After that, studies on DT associated it with ICT-enabled transformation (Wessel et al., 2021). That way, DT started being perceived as an evolution of IT-

enabled transformation since it represented how digital technologies disruptively impacted individuals, organizations, and society (Vial, 2019). Another conceptualization commonly related to DT relates to innovation and disruption, understanding DT as a broader process of change that represents a combined effect that both drives and is driven by innovations and digital disruptions (Skog et al., 2018).

Although DT is a global phenomenon, each country can offer a more or less favorable context for such initiatives to be successful (Limani et al., 2019). In addition, not only nations, but also companies face the challenges of DT, which is dependent of three dimensions: technological, organizational, and social (Švarc et al., 2020). As noticed in previous studies (Vial, 2019; Mergel et al., 2019), DT was primarily related to organizations, and there are significant contrasts throughout definitions concerning the sorts of advancements of technologies encompassed and the nature of the existing transformation (Vial, 2019). In fact, DT is strongly related to organizations, addressing, and emphasizing the impact of ICT in the organizational structure, routines, information flow and capabilities, commonly associated with managerial issues (Vial, 2019; Li, 2020; Stief et al., 2016). Notwithstanding, DT is a process adopted by companies to reach superior performance and sustainable competitive advantage, by converging multiple digital technologies capable of transforming organizational dimensions such as business models, customer experiences and operations configuring a significant change in the ways of working (Ismail et al., n.d., Henriette et al., 2015; Švarc et al., 2020). DT is therefore associated with a profound transformation provided by digital technologies in the activities, processes, and business models of companies, aiming to radically improve their performance (Bekkhus, 2016; Demirkan et al., 2016; Matt et al., 2015).

Moreover, Verhoef et al. (2021) point out three external factors as the main drivers of DT: (1) the growing number of emerging technologies; (2) the drastic change in competition; (3) the change in consumer behavior as a response to the digital revolution. However, despite the growing awareness of the importance of DT in reshaping organizational models and activities, many companies still struggle to assimilate and implement same accordingly (Stief et al., 2016). Becoming digital is a ubiquitous challenge that has been faced by companies in the last decades (Henriette et al., 2015). However, this transformation process is not sustained by technology alone (Nadkarni & Prügl, 2021). Orlikowski (1996) found empirical evidence that organizational transformation was enabled by ICT, but not caused by it. Thus, researchers and organizations gradually have perceived that DT is not only a technological shift (Henriette et al., 2015) – it simultaneously impacts people and networks (Ismail et al., n.d.). Previous studies have identified that successful DT initiatives are associated not only with the use of ICT as some other dimensions not related to technology as culture and leadership (Nadkarni & Prügl, 2021). Besides, both academics and practitioners have recognized the importance of human resources in this endeavor (Gong & Ribiere, 2021) and lack of digital skills has been commonly pointed as a hurdle to the success of DT (Agrawal et al., 2019). Thus, the main critical success factor for DT initiatives is the focus on their transformational dimension rather than on digital issues (Lam & Law, 2019). This involves aligning ICT with people to ensure that the digital potential is associated with qualified professionals, allowing DT to reach the best of its transformative power.

Although DT can be seen, from a business perspective, as a change in how companies use digital technologies to develop new business models, there has not been a multidisciplinary discussion on the subject. Insights about digital business strategy are still underdeveloped, as companies are still dealing with multiple digital shifts within traditional domains (Bharadwaj et al., 2013). While it is difficult for researchers to develop theories, practitioners also face difficulties in developing and implementing DT strategies in their organizations (Gong & Ribiere, 2021). In addition, there are different levels of DT in companies from different countries, although all have gone or will go through the same process (Limani et al., 2019). Thus, it is plausible to realize that a more comprehensive understanding of the concept of DT, considering the context of organizations, is still in progress. Moreover, to understand

how DT unfolds in organizations, it is necessary to assess the organizational readiness for same, namely their digital readiness, e-readiness – or, in a nutshell, their DT readiness.

3. Conceptualizing e-Readiness and Digital Readiness

3.1. e-Readiness

The origin of the e-readiness construct is strongly related to the public dimension, addressing, for example, countries as units (Bui et al., 2003; Dada, 2006; Agrawal et al., 2019). In fact, it stems from an attempt to provide a unified framework for assessing the scope of the digital divide between developed and developing countries (Choucri et al., 2003). In 1998, the Computer Systems Policy Project (CSPP), a public policy group, gathered information from technology companies in the United States and developed an e-readiness assessment tool known as the "Readiness Guide for Living in the Networked World" – the first time the term was applied (Mutula & van Brakel, 2006). However, the main problem with e-readiness is the lack of a common standard of assessment associated with a unified set of measures (Bakry, 2003). Jutla et al. (2002) took the first steps in conceptualizing e-readiness from the competitive perspective of e-business. Besides, from an e-commerce perspective, Bui et al. (2003) developed a framework to assess and monitor the e-readiness index of a country comprising eight factors – digital infrastructure, macroeconomics, investment capacity, education of citizens, competitiveness, access to skilled labor, culture, and cost of living – and 52 related indicators.

In the following years, several studies attempted to critically assess the level of e-readiness in developing countries, such as the studies of Ramayah et al. (2005), Ifinedo (2005), Dada (2006), Mutula & van Brakel (2006), Alghamdi et al. (2011), Al-Eryani & Rashed (2012) and Chipembele & Bwalya (2016), to name just a few. Dada (2006) points out that several governmental and private institutions have formulated e-readiness rankings via different methodologies and divergent definitions of the construct. Thus, the results are not consistent with each other. Berthon et al. (2008) offer a unique perspective on that, presenting a model in which 76% of a country's e-readiness variation is explained by the propensity to corruption and cultural values. Lou & Goulding (2010) also acknowledge the difference in methodologies and the divergent definitions of what is considered "global e-readiness." Besides, they point out that the private sector initially conducted most e-readiness initiatives, as governments used to be late adopters of ICT strategies in general. Thus, very recent studies remain proposing and evaluating e-readiness measurement tools – as an example, a new review was conducted by Pal et al. (2020), focusing on the agricultural sector.

3.2. Digital Readiness

Readiness is related to the condition of being fully prepared or willing to do something (Oxford, 2021). Thus, the concept of readiness can be described as the extent to which an organization is prepared to change its current state to a new desired state (Kaplan & Norton, n.d.). However, digital readiness construct has no universal definition (Nasution et al., 2018), being an emerging construct derived from the concept of change readiness. One of the main contributions of research on digital readiness is the proposition of the criteria for assessing organizational readiness for DT (Giang et al., 2021). Digital readiness, thus, assesses the readiness of an organization before a DT endeavor is undertaken (Pirola et al., 2019).

Although there are some differences, readiness is often used as a synonym for maturity. Maturity models are used to assess readiness, making it possible to identify an organization's position in relation to a specific variable. In fact, they provide frameworks for assessing business performance, enabling comparison and improvement of an organization's level of capability on a given dimension (Becker et al., 2009; Pirola et al., 2019). Digital readiness is also a relevant competitive resource in an innovation economy as it affects the success of national R&D transfer to the entrepreneurial sector (Moeini Gharagozloo et al., 2022). Companies looking forward to implementing DT initiatives must answer key questions about how to get started and which barriers can hamper the success of the DT endeavor

(Jones et al., 2021; Margiono, 2021). These questions are directly related to the readiness of the company to be digitally transformed, focusing on its internal capabilities and barriers (Stoianova et al., 2020).

De Carolis et al. (2017) argue that previous models for measuring digital readiness are incomplete, as they do not consider the complexities of modern businesses. As such, they proposed a maturity model for digital readiness that measures the ability of companies to effectively manage their processes to achieve desired outcomes. However, the model proposed by De Carolis et al. (2017) addresses the manufacturing sector. In the present study, the perspective is more related to the organizational business environment, which considers the human aspect as part of the fundamental characteristics for DT readiness in organizations. Some studies also analyze the relationship between digital readiness and technology acceptance in an individual approach, using the technology acceptance model (TAM) as a parameter (Ashari Nasution et al., 2021; Venkatesh et al., 2003). In this study, the goal is to understand DT organizational readiness as a whole, taking into consideration human aspects (Pirola et al., 2019).

In sum, in terms of conceptualization, the extant scientific literature does not indicate significant differences between e-readiness and digital readiness. Often both constructs are used to define a situation where a certain level of readiness is required to achieve a change from something analogical to something electronic or digital. Indeed, the first studies on e-readiness considered the external environment, usually a country or nation, as the main antecedent for electronic adoption (Choucri et al., 2003). Furthermore, over the years, there are more published articles on e-readiness than on digital readiness - as it will be later demonstrated in this work – as digital readiness is a more recent issue related to the adoption of digital technologies by individuals, organizations, industries, and countries (Nasution et al., 2018). However, both constructs still lack a unified definition (Dada, 2006; Nasution et al., 2018). Based on that, in the extant literature, there is not enough empirical evidence to date on how to design and evaluate a successful DT in organizations, from a business perspective. Therefore, there is a research gap on the exact understanding of organizational readiness for DT (Matt et al., 2015), which underpins the relevance of this work.

4. Methodological Procedures

This study encompasses a literature review that investigates the constructs that comprise organizational digital transformation readiness. The review begins with a selection of articles to identify the works published in related journals, including references that better match the structure and discussion of the principal theme (Webster & Watson, 2002). The data collection was held between April 2021 and May 2022. The review was conducted in chronological order and the delimitation of the period comprised mainly the years 2004 to 2021 – considering that the subject is still recent and practically unexpressed before this period. Articles published in 2022 were not included in the count shown in the graphs, in order to have equal comparison measures, i.e., using only complete years. Notwithstanding, the articles published till May in 2022 were reviewed for the overall analysis and for theoretical purposes. The search aimed to cover articles published in journals associated with the theme in the Web of Science database, as well as in the most important IS journals[1]. The journals investigated were essentially in English, most of which with a double-blind review process. Books, technical reports, and others were excluded.

The first step addressed the search of the following constructs in the Web of Science database, using quotation marks to find the exact match of: “digital transformation”; “e-readiness”; “digital readiness”; and “digital transformation readiness”. The search included the presence of the constructs in the title, abstract and keywords, to analyze and compare the number of papers published per year

[1] One considered the basket of IS top journal according to the Association for Information Systems (AIS) (see https://aisnet.org/page/SeniorScholarBasket).

and in the sum of these years. Albeit there was a search for the total number of articles of the "digital transformation" construct, as the number of articles is much broader, the subsequent analyses were focused on the three other constructs "e-readiness"; "digital readiness"; and "digital transformation readiness". "Digital transformation" was used for the theoretical part, considering the relevance of the journals and the number of citations of the articles.

The second step comprised the analysis of the three subsequent constructs/themes according to the areas of interest in which they were published (categories), allowing the identification of the main categories per theme. This categorization helped to prioritize the themes that are more related to IS, ICT, and other organizational themes. The third step comprised the search for each of the constructs/themes in the related publishers of the most important journals between 2004-2022 in their specific websites, e.g., Elsevier Science Direct, Taylor & Francis. This stage aimed to evaluate the number of publications per theme and to find specific articles more related to DT readiness.

The last step included a deeper analysis of the main articles found, including the most cited and relevant ones, identifying those with scope more related to organizations. The relevant themes related to DT readiness within the articles were then used to build a concept matrix to compare the findings in the literature. Further, a maturity model to assess the organizational DT readiness was proposed. Finally, a framework was developed to assess the organizational readiness for DT by investigating the logical order and connections of categorized themes selected in the reviewed articles. The DT process begins with the definition and assessment of the organizational requirements for its proper digital readiness (Sanchez, 2017).

5. Results

The search in the Web of Science database brought up works published from 2004 to 2021, considering both articles and conference papers. Thus, 5,007 articles related to "digital transformation", 278 articles related to "e-readiness", 91 articles related to "digital readiness" and 7 articles related to "digital innovation readiness" were detected, when analyzing the title, abstract or keywords of the surveyed works, as shown in Figure 1.

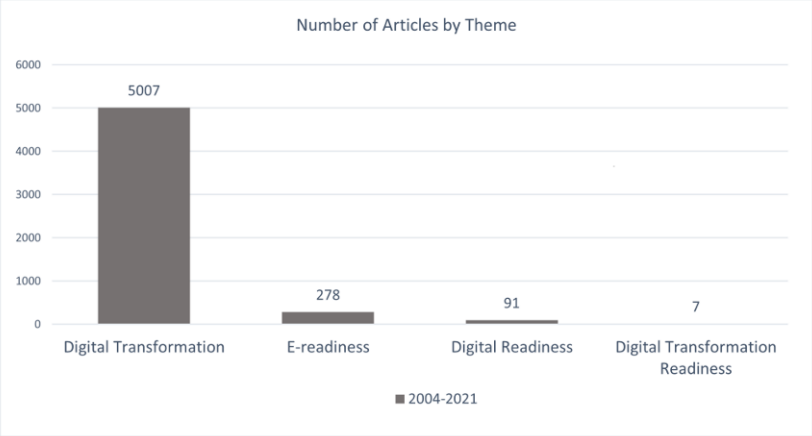


Figure 1. Number of articles by theme 2004-2021.

To focus on the subject of readiness, the theme "digital transformation" was excluded from further analysis due to its broader scope, as it is inserted in sundry academic categories. Figure 2 represents the search conducted, detailing the number of articles published on topics related to readiness, in chronological order and separated by year of publication. E-readiness is by far the most researched topic, totaling 278 articles in the period, covering mainly the governmental and institutional approach to readiness, and mainly addressing the metrics of stages and digital advances in countries. Although the concept "digital readiness" did not have the same numerical representation of "e-

readiness", it had the highest number of publications in a single year, 2021, reaching 36 articles. For all themes, most articles were published from 2017 onwards.

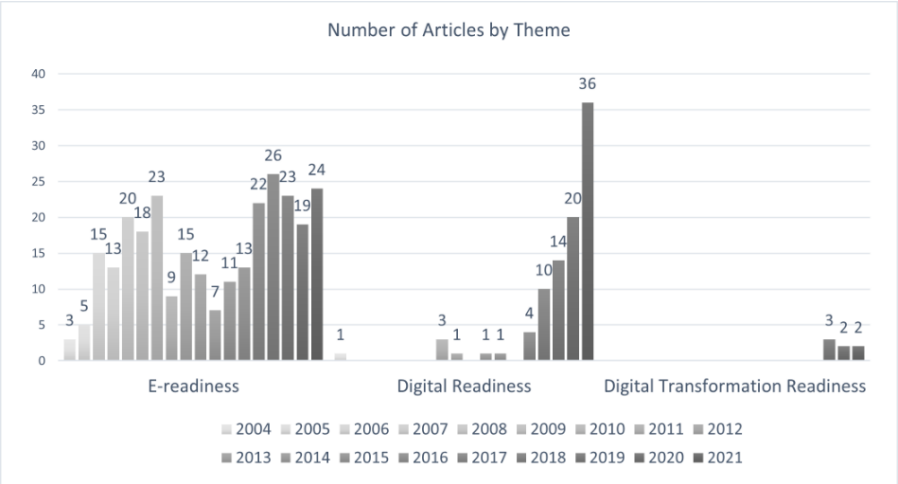


Figure 2. Number of articles by theme per year of publication.

Table I details the number of articles on each topic by Web of Science category in the period 2004-2021. The top twenty categories are listed in that table. It is relevant to mention that themes may be counted in more than one category, i.e., the theme "e-readiness" may be considered in both Business and Management categories so that the total sum of the categories may result in a much larger number than the original count of the articles. For e-readiness, the main categories are in the fields of Computer Science, Information Science, Business, Education, and Management; for digital readiness, the main categories are in the fields of Education, Management, Business, Computer Science, and Economics; and, for DT readiness, the few articles found are divided into the fields of Business, Computer Science, and Economics. In this study, the focus is on articles related to organizational issues, thus covering the areas of Information Systems, Business and Management.

E-readiness		Digital Readiness		Digital Transformation Readiness	
Computer Science Information Systems	54	Education Educational Research	24	Business	2
Information Science Library Science	47	Management	19	Computer Science Theory Methods	2
Business	40	Business	11	Economics	2
Education Educational Research	39	Computer Science Interdisciplinary Applications	8	Automation Control Systems	1
Management	38	Economics	8	Computer Science Interdisciplinary Application	1
Computer Science Theory Methods	35	Engineering Manufacturing	8	Information Science Library Science	1
Computer Science Interdisciplinary Applications	30	Computer Science Theory Methods	7	Management	1
Engineering Electrical Electronic	23	Environmental Sciences	7		
Computer Science Artificial Intelligence	18	Green Sustainable Science Technology	6		
Computer Science Software Engineering	13	Engineering Industrial	5		
Economics	12	Environmental Studies	5		
Telecommunications	12	Information Science Library Science	5		
Multidisciplinary Sciences	11	Computer Science Information Systems	4		
Engineering Industrial	10	Automation Control Systems	3		
Operations Research Management Science	10	Computer Science Artificial Intelligence	3		
Public Administration	9	Computer Science Software Engineering	3		
Engineering Multidisciplinary	8	Energy Fuels	3		
Political Science	7	Engineering Electrical Electronic	3		
Communication	6	Health Care Sciences Services	3		
Social Sciences Interdisciplinary	6	Operations Research Management Science	3		

Table I. Number of articles in main categories per theme.

As presented in Table II, a complementary online search conducted on the website of main publishers from 2004-2022 nominated the journals with the higher number of articles, resulting in 264 papers on "e-readiness", 72 articles on "digital transformation", and 8 articles on "digital transformation readiness". The table is ranked from the highest to the lowest number of articles found, ordered first by the "e-readiness" results, then by the subsequent two themes. Some of the main journals in the Information Systems and ICT subjects were marked in gray. Information Systems

Journal (ISJ), Information Systems Research (ISR) and Management Information Systems Quarterly (MISQ) do not present any article related to any of the three abovementioned themes.

	E-readiness	Digital Readiness	Digital Transformation Readiness
The Electronic Journal of Information Systems in Developing Countries	48		
Government Information Quarterly	28	3	
Information Technology for Development	25	1	
International Journal of Information Management	13	3	
Journal of Global Information Technology Management	12		
Telecommunications Policy	11	4	
The Information Society	11	1	
International Journal of Electronic Commerce	10		
Telematics and Informatics	9		
IFAC-PapersOnLine	8	4	1
Procedia Computer Science	6	8	1
Industrial Marketing Management	6	3	
Electronic Commerce Research and Applications	6	2	
International Journal of Network Management	6		
Journal of Information Technology & Politics	6		
Information & Management	5		
International Information & Library Review	5		
International Journal of Public Administration	5		
Journal of Internet Commerce	5		
Technology in Society	4	2	
European Journal of Information Systems (EJIS)	4	1	
Cogent Business & Management	4		
Journal of Business Research	3	9	
Journal of Information Technology (JIT)	3		
Electronic Markets	3		
Journal of Organizational Computing and Electronic Commerce	3		
Journal of Promotion Management	3		
Journal of Urban Technology	3		
Technovation	2	2	
International Journal of Project Management	2		
Journal of Strategic Information Systems (JSIS)	1	1	
Journal of the AIS (JAIS)	1		
Information Systems Management	1		
Journal of Marketing Management	1		
Technology Analysis & Strategic Management	1		
Procedia Manufacturing		10	
Technological Forecasting and Social Change		6	
Journal of the Association for Information Science and Technology		3	
Journal of Manufacturing Systems		2	1
Computers & Industrial Engineering		2	
Computers in Industry		2	
Journal of Purchasing and Supply Management		2	
Journal of Management Information Systems (JMIS)		1	
Future Generation Computer Systems			1
Sustainable Production and Consumption			1
The Journal of Academic Librarianship			1
Journal of Eastern European and Central Asian Research			1
Information Systems Journal (ISJ)			
Information Systems Research (ISR)			
Management Information Systems Quarterly (MISQ)			
Total	264	72	7

Table II. Number of articles per theme by related journals.

After exploring the databases, a selection of the articles was held, prioritizing the ones addressing the "digital transformation readiness" (DTR) and "digital readiness" (DR) themes. The articles chosen were those most related to organizational/business aspects and with the highest number of citations. Thus, Table III presents the matrix of concepts, with the articles and the main constructs related to ODTR, which will be used to develop a framework on this subject. The constructs were also combined according to semantic organization related to digital business readiness and DT in organizations. Table III presents the articles in chronological order and according to the number of related constructs found in each article, from the highest to lowest one. Besides, the oldest article on the subject is from 2017, confirming, thus, the novelty of the theme. The following constructs were found, namely Technology, Change, Innovation, Disruption, Assessment, Maturity, Performance, Processes, Deployment, Implementation, Organization, Company, Small and Medium Enterprises

(SME), Management, Strategy; Resources, Capabilities, Competitiveness, Business, Business Model, Culture, Environment; Human (Aspects), and Sustainability.

		Technology	Change, Innovation, Disruption	Assessment, Maturity, Performance	Processes, Deployment, Implementation	Organization, Company, SME	Management, Strategy	Resources, Capabilities	Competitiveness	Business, Business Model	Culture, Environment	Human (aspects)	Sustainability
De Carolis et al. (2017)	DR	x	x	x	x	x	x	x	x	x	x	x	x
Alharbi (2019)	DTR	x	x	x	x	x	x		x	x			x
Lam & Law (2019)	DR	x	x	x	x	x	x	x	x	x	x		x
Limani et al. (2019)	DTR	x	x	x	x	x	x	x	x	x			
Machado et al. (2019)	DR	x	x	x	x	x	x	x	x	x	x		x
Pirola, Cimini & Pinto (2019)	DR	x	x	x	x	x	x	x		x	x	x	
Pramanik, Kirtania & Pani (2019)	DTR	x	x	x	x	x	x	x	x	x	x	x	
Bican & Brem (2020)	DR	x	x	x	x	x	x	x	x	x	x		
Caiado et al. (2020)	DR	x	x	x	x	x	x	x	x	x	x	x	
Genest & Gamache (2020)	DR	x	x	x	x	x	x	x	x	x	x	x	x
Gfrerer et al. (2021)	DR	x	x	x	x	x	x	x	x	x	x	x	
Stoianova, Lezina & Ivanova (2020)	DTR	x	x	x	x	x	x	x	x	x	x	x	
Svarc, Laznjak & Dabic (2020)	DTR	x	x	x	x	x	x	x	x	x	x	x	
Deja, Rak & Bell (2021)	DTR	x	x	x	x	x	x	x	x		x		
Hoa & Tuyen (2021)	DTR	x	x	x	x	x	x	x	x	x	x	x	x
Criado-Perez et al. (2022)	DTR	x	x	x	x	x	x	x	x	x	x		x

Table III. Concept Matrix for "Digital Readiness" (DR) and "Digital Transformation Readiness" (DTR)

6. Organizational Digital Transformation Readiness (ODTR)

The literature review lists the main characteristics of the constructs that make up organizational digital transformation readiness. Through an extensive search of terms related to e-readiness, digital readiness and DT readiness, it was possible to verify that research on the latter subject is still scarce. Even in the organizational area, several articles are related to aspects of Industry 4.0 and manufacturing processes, leaving aside other business aspects. Thus, given the importance of the transformational processes that companies have been facing, with regard to the growing adoption of digital technologies, studies related to the subject under analysis are essential. However, before defining the architecture of the aforementioned framework, it is essential to identify the key organizational processes for companies to achieve DT, as these business-related processes must portray the company's strategy (Jones et al., 2021).

Before designing the ODTR framework, it was necessary to define the maturity levels that would enable the company's readiness to change. The conceptualization of "maturity" is related to an organized set of levels to assess management practices in different capabilities or process areas in a qualitative or a semi-quantitative way (Macchi & Fumagalli, 2013; Team, 2002). Maturity models can be applied in different specific scopes, enabling a general understanding of what has to be done for developing maturity assessment (Macchi & Fumagalli, 2013). The levels proposed in this study were inspired by the CMMI framework (Capability Maturity Model Integration), which has its roots in the software engineering area - having been generalized to incorporate other areas - as well as the DREAMY (Digital REadiness Assessment MaturitY model) (Macchi & Fumagalli, 2013; De Carolis et al., 2017). The decision to use the CMMI framework was based on the clarity, flexibility, and well-designed model, which provides a structure that helps organizations appraising their maturity, describing discrete levels of process improvement within each process area (Team, 2002). In this work, the CMMI maturity model was used (De Carolis et al., 2017) with some adjustments made to fit organizational digital readiness assumptions in order to consider socioeconomic differences - as many

businesses in developing countries would start at level 0, rather than level 1, as some models propose, as detailed later.

Figure 3 describes the maturity levels for ODTR, according to the CMMI model (Team, 2002) and considering adaptions provided by De Carolis et al. (2017), as abovementioned. In ***Level 0*** (Incomplete), the process is unpredictable, poorly controlled, or without any control; the management level is reactive and with inconsistent performance. ***Level 1*** (Initial) embraces an initial approach to meeting the objectives of the practice area, beginning to address performance issues. ***Level 2*** (Managed) includes the practices from Level 1, establishing a complete set of practices that fully address the intent of the practice area—requiring no specific assets and identifying and monitoring the progress. ***Level 3*** (Defined) incorporates Level 2 practices, adding organizational standards to address specific characteristics, using and contributing to organizational assets, and focusing on achieving performances. ***Level 4*** (Integrated and Interoperable) incorporates Level 3 practices, with integration and interoperability of applications, as well as information exchange, being thoroughly planned and implemented considering enabling technologies, uses quantitative techniques to understand performance and achieve quality; finally, ***Level 5*** (Digital Oriented) builds on level 4 practices with a fully digital oriented process, based on a robust technological infrastructure and a high potential organizational growth, providing fast and secure information exchange, in collaboration among the company functions and the decision making.

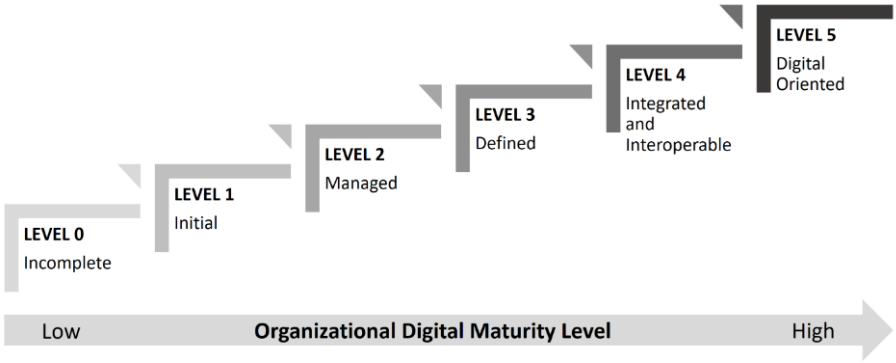


Figure 3. Maturity levels for Organizational Digital Transformation Readiness. Adapted from Team (2018, p. 13) and De Carolis et al. (2017, p. 17).

Differently from the DREAMY approach, presented by De Carolis et al. (2017), as well as other models in the literature that prioritize relevant stages for manufacturing processes (Machado et al., 2019; Gfrerer et al., 2021), this study focus on the organizational approach, also considering human aspects as essential to enable ODTR. Combining the most relevant dimensions associated with organizational resources, capabilities and managerial decisions involved in the digital readiness process in the literature, the main criteria were selected vis-à-vis the key dimensions for ODTR, as detailed in Table IV.

Dimensions	Criteria
Technological Resources	- Technology availability (infrastructure to support digital processes) - Use of BI, data-driven, digital tools for management - Innovative TR
Business Processes	- Processes to support digital businesses - Evaluation processes to guarantee digital evolution of businesses
Management Capability	- Management and Leadership Skills on digital environment - Develop and implement digital environment and businesses as core strategy - Innovation driven mindset - Ability to evaluate performance indicators to provide sustainable businesses
Human Capability	- Human digital skills development - Ability to work with digital resources - Understand and disseminate digital culture
Corporate Culture	- Culture of digital environment - Competitive awareness - Sustainable digital business culture - Corporate institutional communication to disseminate digital culture

Table IV. Dimensions and Criteria associated with Organizational Digital Transformation Readiness (ODTR)

Those dimensions encompass the indispensable resources and capabilities related to organizational digital readiness, representing the success factors for DT endeavors. Each dimension and criterium must be evaluated according to the previous maturity levels proposed to achieve digital readiness. The dimensions and the subsequent criteria are:

Technological Resources comprise available technology (infrastructure to support digital processes), use of business intelligence (BI) and other data-oriented digital management tools, as well as innovative IT solutions such as cloud computing, big data, internet of things (IoT), virtual/augmented reality, robotics, artificial intelligence (AI), machine learning (deep), etc. (De Carolis et al., 2017; Pramanik et al., 2019; Stoianova et al., 2020; Hoa & Tuyen, 2021).

Business Processes encompass all the processes needed to support digital businesses. DT requires the change of organizational processes and/or development of new business models (Reis et al., 2018). The processes are carried out via digital platforms and consist of well-designed and described formal procedures that may be divided into several sub-processes. Besides, processes must be constantly evaluated to guarantee the digital evolution of the business (Pramanik et al., 2019; Stoianova et al., 2020; Bican & Brem, 2020).

Management Capability encompasses management and leadership skills in the digital environment. This includes the development and implementation of a digital environment and definition of a digital business model as a core strategy. Managers must also have an innovation-driven mindset, capable of perceiving the digital changes in the competitive environment. This dimension also refers to the managerial ability to evaluate performance indicators to provide a sustainable digital business (Stoianova et al., 2020; Hoa & Tuyen, 2021; Gfrerer et al., 2021).

Human Capability is related to the development of digital skills of personnel directly or indirectly related to tasks that demand digital knowledge. These skills are usually verified in hiring processes or can be developed through specific training programs. Not only employees, but also third parties must have the ability to work with digital resources. Another aspect of human capability is the understanding and dissemination of the organization's digital culture (Pramanik et al., 2019; Pirola et al., 2019; Hoa & Tuyen, 2021; Gfrerer et al., 2021).

Corporate Culture encompasses the development and perpetuation of a digital environment throughout all corporate levels by means of institutional communication and actions. As organizations understand the need to perform in a digital competitive scenario, they seek to promote a sustainable

digital culture, namely an innovative culture. Stoianova et al. (2020) merged corporate culture with human resources in one dimension. However, corporate culture is a broader concept, integrating human resources, processes, management, physical organizational aspects, and technology (Gfrerer et al., 2021; Stoianova et al., 2020; Hoa & Tuyen, 2021).

6.1. Proposed Framework

From the theoretical background presented and the analysis of the key constructs and organizational dimensions related to digital readiness, the Organizational Digital Transformation Readiness (ODTR) framework is proposed, as detailed in Figure 4. The framework proposes a logical and structured organization for ODTR. The framework starts at the broadest level, representing the market and context in which the company operates, called the Competitive Environment. The second level – the focus of the study – is the organization, including the resources and capabilities or key dimensions for DT readiness, namely Technological Resources, Business Processes, Capability Management, Human Capability and Corporate Culture, previously detailed. These key dimensions do not follow a pre-defined hierarchy or order, interacting with each other and being driven by top management.

As mentioned earlier, each level needs to reach the required maturity to achieve digital readiness. At the organizational level, once the levels of maturity of the five key dimensions have been reached, it is up to top management to propose, develop and implement a digital strategy that translates into a digital business model. This business model will interact with all processes and digital resources previously listed, and its design will be determined by the level of digital innovation intended by the organization in its competitive environment. It is important to emphasize that, over the process, the DT readiness indicators must be monitored and controlled, until all levels for a complete DT are reached. Therefore, DT will happen when the necessary readiness is achieved. Once the transformation occurs, it consequently impacts back the organization (as indicated by the arrow from the DT box towards the organization box). Besides, once the organization is transformed, it also impacts the competitive environment in which it is inserted (as indicated by the arrow from the DT box to the competitive environment box). Given the constant technological changes organizations have been witnessing, the framework is a cyclical process.

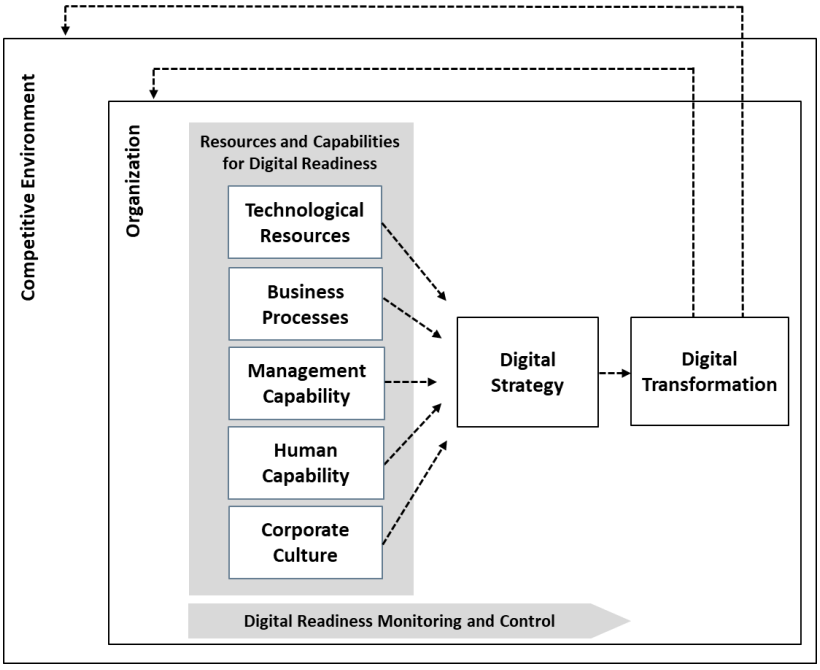


Figure 4. Organizational Digital Transformation Readiness (ODTR) Framework. Property of the authors.

When assessing the DT readiness of companies in different socioeconomic realities, through the ODTR framework, three propositions can be formulated:

P1. Successful DT initiatives in organizations depend directly on their digital readiness, measured through their level of maturity in the ensuing dimensions: technological resources, business processes, management capacity, human capacity, and corporate culture (De Carolis et al., 2017; Isaev et al., 2018; Pirola et al., 2019; Pramanik et al., 2019; Stoianova et al., 2020; Giang et al., 2021).

P2. The readiness level of firms for DT has a direct impact on the conceptualization, design, and implementation of their digital strategies (Isaev et al., 2018; Pramanik et al., 2019; Bican & Brem, 2020).

P3. The level of maturity for DT of a business impacts directly its level of competitiveness in the market in which it operates (Pramanik et al., 2019; Stoianova et al.

7. Conclusions

This study proposed an integrative framework for assessing the ODTR based on a review and analysis of the extant scientific literature on this subject. After completing that task, this paper revealed an overlap between the definitions of e-readiness and digital readiness constructs. Indeed, the extant scientific literature does not demonstrate so far significant distinction between e-readiness and digital readiness, as both are supposed to indicate the level of maturity required from a country, institution, organization, or person to be digitally transformed successfully. This work also unveiled the lack of a unified and monosemic definition for the "digital transformation readiness" construct, especially in the organizational realm. In the universe of articles retrieved from the Web of Science (2022) database, e-readiness represents about 74% of the works, digital readiness represents 24% and digital transform readiness represents only 2% of the articles - most of them published after 2017.

Indeed, ODTR is a relatively new construct in the academic field, reporting the idea of measuring the level of readiness of an organization for ICT implementation and use, combined with other organizational issues, to enable a successful DT of same. In fact, studies on organizational readiness for DT are also quite scarce, representing this fact a research opportunity. In short, there is an academic gap associated with the investigation of the level of maturity required for the various dimensions of an organization for the success of DT initiatives carried out in it, as corroborated by the literature review carried out. This academic gap ranges from the assessment of the existing context for organizational DT initiatives to their implementation and final assessment.

It is also noteworthy that there are several relevant concepts involved with organizational readiness for DT, as already mentioned, which is corroborated by the evidence found in the related literature. These concepts are related not only to ICT, but also to several other organizational dimensions, such as resources, processes, organizational culture and environment, competitiveness, management, human competencies, business models, sustainability, among others, thus confirming the multidisciplinarity (and even the interdisciplinarity) of the theme.

7.1. Theoretical Contributions

This study aimed to provide a comprehensive and in-depth understanding of the concepts and methodological aspects that cover the topic of ODTR. The literature review identified the main constructs related to digital readiness and DT readiness, as well as the logical configuration of the critical factors associated with successful DT initiatives in organizations, which allowed the formulation of an exploratory framework on it. The proposed framework articulates the main dimensions and criteria necessary to measure and evaluate the organizational readiness for DT. It also provides a straightforward visualization of the essential steps required in an organization to develop a business model associated with its DT. The framework also includes a competitive environment

perspective, proposing that the DT process is cyclical: once the transformation is accomplished in the organization, it impacts back the competitive environment in which the organization operates.

Another contribution of this work is the proposal of a maturity model to assess organizational digital readiness. The model allows for conceptual adjustments that suit organizational assumptions and sundry socioeconomic scenarios. In addition to the theoretical contribution associated with the proposition of the ODTR framework, this work aims to be a guide for practitioners to develop and evaluate DT efforts in their companies. Finally, the study aims to contribute to the integration of diverse and multifaceted fields of knowledge associated with DT, such as ICT and Organizational Studies, to name just a few.

7.2. Limitations and Further Steps

While this review is not exhaustive, the proposed framework is intended to provide a broader understanding of the steps an organization must take to succeed in a DT initiative in a fast-changing digital-dependent world. However, some limitations must be considered. The search was limited to the Web of Science database and sites from main publishers in Information Systems and Management. Although this allowed a broad search, some publications may not have been considered. In addition, some limitations were established regarding the scope and search criteria.

Besides, future steps may include further elaboration and validation of the proposed model, with a broader investigation on how organizations have responded to the challenge of DT and their readiness in practice to do so (Soto-Acosta, 2020). Moreover, comparative studies might also incorporate specific contextual aspects associated with distinct business industries in different socioeconomic contexts, such as ODTR in developing countries (Raghavan et al., 2021). In addition, further investigations can strengthen the understanding of how (and by whom) some relevant decisions about DT initiatives are made in organizations (Larjovuori et al., 2018). Finally, future research may amplify the current debate about new digital business models (Bican & Brem, 2020) associated with DT.

Overall, with the accelerated pace of digital technologies and the profound changes that the world has faced lately, expanding the theoretical scope associated with the impact of emerging ICT on organizations is essential to monitor and evaluate the latest initiatives on DT, as well as to predict the future changes on same. This work, therefore, intends to move in this direction, by proposing a new way of conceptualizing and evaluating the processes behind the DT initiatives that organizations are currently experiencing.

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6. Paper II

CRITICAL SUCCESS FACTORS FOR ORGANIZATIONAL DIGITAL TRANSFORMATION: A DELPHI STUDY IN THE RETAIL INDUSTRY

Abstract

Digital Transformation is a multifaceted concept related to profound changes in society and contemporary organizations, brought about by innovations enabled by the implementation of emerging Information and Communication Technologies (ICT). As such, the integration of ICT with organizational structures, processes, and outcomes has garnered considerable attention from researchers, particularly within the context of digital transformation. However, empirical evidence on the critical factors for successful digital transformation initiatives in organizations remains scant. Thus, this study aims to investigate the key success factors that enable digital transformation in organizations, by ranking their degree of importance via a Delphi-based approach. The focus of the study is the retail sector, which has substantial socioeconomic influence, has undergone notable transformations in recent times, and remains an understudied industry. The final number of subjects consisted of 52 respondents, and the results show that most of the critical success factors in organizational digital transformation are related to management capability, capacity, and business processes, not to technology autonomously. Furthermore, leadership in promoting new ICT and fostering innovation, as well as constant assessment of organizational digital evolution in line with market trends, also proved to be vital for the success of digital transformation initiatives in enterprises.

Keywords: Digital Transformation, Organizational Digital Transformation, Critical Success Factors, ICT, Retail, Delphi.

1. Introduction

The intersection of technology with organizational structures, processes and outcomes has been a focal point for researchers for quite some time (Orlikowski, 2000). Information and Communication Technology (ICT) rises in modern society as a facilitator of economic development through active participation in the globalized competition (Avgerou, 2010) in a way that significant advancements in ICT have made available several functionalities that have allowed firms to adapt their business infrastructure to completely new digital era (Bharadwaj et al., 2013). Due to that, the concept of digital transformation (DT) has emerged in academia and among practitioners (Wessel et al., 2021) as a social construction that implies a cultural transformation since it represents a process of social interaction and collective learning, by which citizens embrace and disseminate digital technologies (Švarc et al., 2020). The sociotechnical approach, prevalent in analyzing technological advancements within organizations, posits that organizational transformation is facilitated by technology, albeit not instigated by it, challenging the notion of technology as an external force, and thus highlighting its entanglement with the social context (Orlikowski, 1996). Corroborating this perception, Lam & Law (2019) put forward that a paramount factor for success in DT initiatives is the focus on their transformational dimension and not solely on digital issues.

However, although DT has emerged as a phenomenon in the area of Information Systems (IS) in recent years, it still does not have a unified definition in the existing literature and is often seen simplistically as a continuous process of emerging technology adoption for significant change (Vial, 2019; Teichert, 2019). Thus, there continues to be a lack of empirical evidence for the design, implementation and evaluation of successful DT initiatives within organizations (Stief et al., 2016; Vial, 2019; Lanzolla et al., 2020), particularly when considering the perspective of an organization rather than just the industry in which it operates.

Besides, the notion of DT has become ubiquitous and is often understood as a change in how a company employs digital technologies to develop new digital business models that helps same generating more value (Lanzolla et al., 2020). Organizations have long confronted the challenge of transitioning to digital formats, a dilemma that has persisted over recent decades (Henriette et al., 2016). Indeed, the digital revolution has transformed the nature of many organizations in both public and private domains by radically changing processes and business models in almost all industries (Demirkan et al., 2016) and DT has been an imperative for organizations to remain competitive in the current digital world (Vial, 2019). That way, organizations are responding to the challenges of competition and globalization set by the fast and deep changes arising from digital technologies (Reis et al., 2018; Nasution et al., 2018).

Despite what has been presented, to date there is a research gap with regard to understanding the dimensions involved in DT, as well as the factors necessary for its successful implementation in organizations (Henriette et al., 2016; Matt et al., 2015; Michelotto & Joia, 2021), which supports the relevance of this work. Thus, to contribute to the extant literature on the subject, this article aims to investigate the main drivers for successful DT in organizations. For this endeavor, the retail sector was selected due to its significant socioeconomic impact and the notable transformations, particularly digital, that it has undergone in recent years.

This work has the following structure besides this section: section 2 describes the theoretical background used; section 3 details the methodological approach adopted; section 4 presents the results; and section 5 discusses them to generate conclusions - which include the main contributions of this article, its limitations and future steps to be followed.

2. Theoretical Background

2.1 Digital Transformation in Organizations

As society has evolved with the fast-growing use of digital technologies, the DT construct has gained multifaceted approaches in the literature as well as several definitions (Henriette et al., 2016). In fact, as a buzzword, DT has had its credibility undermined, which has caused turbulence in both academic and professional fields (Gong & Ribiere, 2021; Verina & Titko, 2019). The literature on DT is quite diverse, involving different clusters of knowledge, and its definition is not yet consolidated (Nadkarni & Prügl, 2021). As DT has different meanings for different audiences, this concept has not been equably applied in the IS literature, being so associated with different perspectives (Voß & Pawlowski, 2019). Probably one of the most comprehensive works on the concept of DT was carried out by Vial (2019, p. 118), who defined DT as "a process that aims to improve an entity by triggering significant changes in its properties through combinations of information, computing, communication, and connectivity technologies." Another conceptualization commonly related to DT is related to innovation and disruption, understanding digital transformation as a broader process of change, representing a combined effect that both drives and is driven by innovations and digital disruptions (Skog et al., 2018). Additionally, Verhoef et al. (2021) highlight three external catalysts for DT: (1) the proliferation of emerging technologies; (2) substantial shifts in competitive landscapes; (3) changes in consumer behavior as a reaction to the digital revolution.

Fitzgerald et al. (2014 p.2) highlight sundry pertinent technologies within their definition of DT, namely "the utilization of novel digital technologies (such as social media, mobile, analytics, or embedded devices) to facilitate significant business enhancements (such as improving customer experience, streamlining operations, or innovating new business models)." However, prominent theories in social sciences have cautioned against oversimplifying DT in organizations as solely the adoption of ICT. Particularly in the study of ICT and society, including organizational studies, two convergent approaches help the understanding of this relationship, namely, Actor-Network Theory (ANT) and Sociomateriality. ANT is a theoretical framework that focuses on understanding social phenomena by examining the relationships and interactions between human actors (individuals or groups) and non-human entities such as objects, technologies, institutions (Callon, 1999; Latour, 2005). Sociomateriality, on the other hand, recognizes that social practices are intertwined with material artifacts, technologies and infrastructures, and these material elements not only shape social practices but are also shaped by them (Orlikowski & Scott, 2008; Leonardi, 2013). Thus, ICT on its own does not sustain the entire transformation process (Nadkarni & Prügl, 2021). In this sense, researchers found empirical evidence that organizational transformation was enabled by ICT, but not caused by it, and gradually perceived that DT was not only a technological shift as it simultaneously impacts people and networks (Orlikowski, 1996; Henriette et al., 2016; Ismail et al., 2017).

Various academic perspectives on ICT have evolved alongside organizations, as both ICT and organizations are undergoing profound transformations, revealing unprecedented potential derived from the synergy between the two (Orlikowski, 2000). Indeed, organizations have been facing the pervasive dilemma of becoming digital in the last decades (Henriette et al., 2016). Initially, studies were related to the strategic role that IT used to play to support business strategies (Henderson & Venkatraman, 1992). Then, DT was mainly associated with IT-enabled transformation (Wessel et al., 2021) and regarded as an evolution of IT-enabled transformation, since it better represented how digital technologies disruptively impacted individuals, organizations and society (Vial, 2019). In fact, DT encompasses adopting digital technologies to revolutionize business operations, processes and business models, thereby realizing organizational objectives (Frey et al., 2023). It has evolved into an indispensable facet of contemporary organizations, affording them the adaptability required to thrive amidst the dynamic landscape of business (Westerman et al., 2011; Kane et al., 2021). Kane et al. (2015) also suggest that DT strategy has its power in its scope and objectives rather than in technology.

A salient dimension of DT comprises its influence on the trajectories of value creation, notably, the reconfiguration of business models within organizations (Vial, 2019). The business infrastructure has been transformed into digital with strengthened interconnectivity among products, processes and services (Bharadwaj et al., 2013). Scholarly literature advocates the adoption of more flexible organizational structures, the incorporation of agile organizational forms, and the establishment of digital functional areas to engender an environment conducive to innovation, thereby facilitating swift responses to digital disruption (Volberda et al., 2021; Verhoef et al., 2021; Maedche, 2016; Gerster et al., 2020). DT involves the implementation of digital resources to support the transformation of business models into digital models associated with a new networked value chain, which can impact the entire organization, thereby configuring a significant change in the ways of working (Henriette et al., 2016; Švarc et al., 2020). The digital revolution is providing means for improving efficiency, effectiveness, sustainability and innovation, by transforming the nature of many organizations and revitalizing the business models of various sectors, such as healthcare, finance, education, retail, tourism, transportation, telecommunications, energy, agriculture, government, and many others (Demirkan et al., 2016).

As observed in previous studies, DT has been applied mainly to organizations, addressing and emphasizing the impact of ICT on organizational structure, routines, information flow and organizational capabilities (Vial, 2019; Mergel et al., 2019; Li, 2020). DT is, therefore, considered a

profound transformation provided by digital technologies in companies' activities, processes and business models to leverage the changes and opportunities to radically improve their performance at different levels in areas such as products, organizational structures and/or automation (Demirkan et al., 2016; Bekkhus, 2016; Matt et al., 2015). DT is also a dynamic process that encompasses the connections between and within organizations, requiring same o be prepared to implement action plans grounded in technological possibilities (Westerman, 2018).

In short, DT is related to the process adopted by companies to achieve superior performance and sustainable competitive advantage, through the convergence of multiple digital technologies capable of transforming sundry organizational dimensions, such as business models, customer experience and operations (Ismail et al., 2017).

2.2. Critical Success Factors for DT in Organizations

Due to the significant shifts in transformation processes that companies are undergoing with the widespread integration of digital technologies, it is imperative to delve into studies focused on this subject. It is crucial to pinpoint the organizational processes and its structuring impacts that play a pivotal role in realizing DT (Henriette et al., 2016; Reis et al, 2018). These processes, closely tied to the company's digital strategy, must be clearly identified (Weill & Woerner, 2018).

In the academic domain, the notion of Critical Success Factors (CSFs) is expounded as the constrained set of domains within which the achievement of favorable outcomes guarantees the competitive prosperity of the organization (Dinter, 2013). Boynton & Zmud (1984) offer a widely accepted definition of CSFs, describing them as the few essential elements that must perform effectively to guarantee success of a manager or an organization. CSF theory works as an evaluative tool to assess an organization's performance in achieving its objectives. That way, Shankar et al., (2018) have utilized the CSF theory as a mechanism to validate success factors within the context of DT.

According to Henriette et al. (2016), organizational digital capabilities are pivotal to the success of DT initiatives. To this end, leaders must redefine their roles and responsibilities to align with the goals of the transformation process and must instill urgency to drive change and transformation within the organization (Hai et al., 2021). Additionally, organizations must define clear workforce strategies, based on digital skills and capabilities, to achieve future objectives (Hai et al., 2021). Holotiuk & Beimborn (2017) proposed forty critical success factors for a digital business strategy (DBS) grouped into eight dimensions, namely the Organization itself, Sales and Customer Experience, Culture and Leadership, Capabilities and HR Competencies, Foresight and Vision, Data and IT, Operations, and Partners. Bharadwaj et al. (2013) also synthesize key themes and their respective success metrics to guide the academic debate towards digital businesses, namely: Scope of Digital Business Strategy, Scale of Digital Business Strategy, Speed of Decision Making, and Sources of Value Creation and Capture.

The literature review presented in the first paper of the dissertation (Michelotto, 2023) generated a framework addressing the key dimensions for organizational DT readiness (Figure 1). These dimensions incorporate the resources and capabilities that are integrated into the main organizational processes that lead to DT.

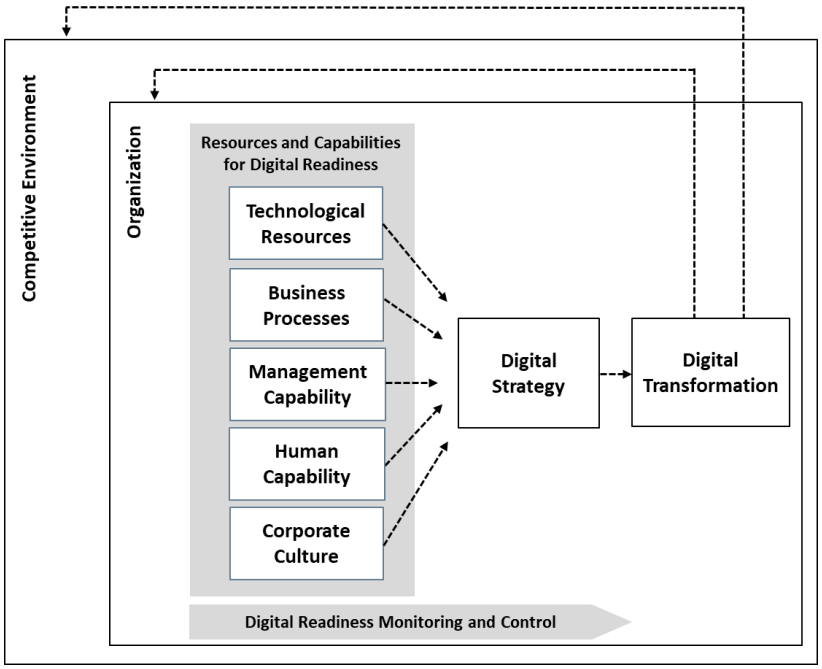


Figure 1. Organizational Digital Transformation Readiness Framework (Michelotto & Joia, 2024)

Those dimensions represent the resources and capabilities that enable organizations to be ready to be digitally transformed, comprising the core structure for the critical success factors for DT, as described below.

D1. Technological Resources relate to all technological infrastructure that support digital processes in organizations such as use of business intelligence (BI) and other digital management tools, all digital data repository or collecting process, as well as digital innovations such as cloud computing, big data, internet of things (IoT), augmented/virtual reality artificial intelligence (AI), machine learning and, more recently, the use of generative pre-trained transformers (GPT). Based on this, the following critical factors for the success of DT in organizations are set forth:

- CF1. Technological infrastructure to support the business, such as Business Intelligence (BI) tools, Cloud, other digital automation, digitization, and management tools (Fitzgerald et al., 2014; Agrawal et al., 2019; Hoa & Tuyen, 2019; Stoianova et al., 2020; Bican & Brem, 2020; Gong & Ribiere, 2021; Wengler et al., 2021).
- CF2. Use of e-commerce platforms (e-commerce sites, sales apps, or other mobile businesses) (Fitzgerald et al., 2014; Hansen & Sia, 2015).
- CF3. Use of the latest innovative technologies such as Artificial Intelligence (AI), machine learning, augmented reality, robots, and others (Fitzgerald et al., 2014; Agrawal et al., 2019; Pramanik et al., 2019; Hoa & Tuyen, 2021; Ghobakhloo & Iranmanesh, 2021).

D2. Business Processes encompass all the processes needed to strengthen digital businesses. This evolution requires adjustments to organizational processes and/or the creation of new business models in order to thrive in the digital landscape (Reis et al., 2018). As business strategy moves through DT, companies across industries must contemplate the importance of network effects and multilateral business models (Bharadwaj et al., 2013). The processes are carried out via digital platforms and must be constantly evaluated to guarantee the digital evolution of businesses. They consist of well-designed and described formal procedures that cover aspects related to communication and social media. Therefore, the corresponding critical factors for the success of DT in organizations comprehend:

- CF4. Constant assessment of the company's digital evolution and identification of opportunities that could lead to new digital business models (Fitzgerald et al., 2014; Bharadwaj et al., 2013; Mhlungu et al., 2019; Bican & Brem, 2020).
- CF5. Communication processes in digital businesses, such as acting on social media to create value and engagement with consumers (Fitzgerald et al., 2014; Bharadwaj et al., 2013; Mhlungu et al., 2019; Pelletier & Cloutier, 2019; Gfrerer et al., 2021).
- CF6. Multiple sales channels, such as physical stores, online stores, and social media (Hansen & Sia, 2015; Mhlungu et al., 2019).

D3. Management Capability extends to both management and leadership skills, particularly within the digital realm. This entails skills to develop and execute digital environments, establishing digital business models as a fundamental strategic component. Additionally, managers need to cultivate an innovative mindset to navigate the evolving digital landscape in highly competitive environments. They must possess managerial capabilities to assess performance metrics effectively, ensuring the establishment of sustainable digital enterprises. This dimension leads to the ensuing critical factors for the success of DT in organizations:

- CF7. Leaders oriented towards promoting the implementation of new technologies and open to innovative ideas and actions (Fitzgerald et al., 2014; Agrawal et al., 2019; Pirola et al., 2019; Stoianova et al., 2020; Hoa & Tuyen, 2021; Gfrerer et al., 2021; Wengler et al., 2021).
- CF8. Leaders with knowledge of new technologies and their application in their businesses (Hansen & Sia, 2015; Mhlungu et al., 2019; Stoianova et al., 2020; Hoa & Tuyen, 2021; Gfrerer et al., 2021; Wengler et al., 2021).

D4. Human Resources and Capability lies on enhancing the digital competencies of employees and third parties engaged in roles requiring digital expertise, whether directly or indirectly. The human component is pivotal in processes leading to DT in companies (Gong & Ribiere, 2021) and the lack of digital skills is commonly pointed as an important hurdle to the success of DT (Agrawal, Narain & Ullah, 2019). Digital skills are either typically assessed during the recruitment phase - job candidates may already possess them - or developed through specialized training initiatives provided by the companies throughout the careers of current employees. Furthermore, the human component is essential to understand and disseminate the digital culture within the organization, leading to the following critical factors (Pramanik et al., 2019; Pirola et al., 2019; Hoa & Tuyen, 2021; Gfrerer et al., 2021).

- CF9. Employees with the ability to work with digital resources Agrawal et al., 2019; Pirola et al., 2019; Gfrerer et al., 2021; Stoianova et al., 2020; Wengler et al., 2021).
- CF10. Training and skills development programs for working with digital resources (Pirola et al., 2019; Stoianova et al., 2020; Gfrerer et al., 2021).

D5. Corporate Culture relates to fostering and sustaining the digital environment across all levels of the organization through communication and institutional initiatives. Although Stoianova et al. (2020) group corporate culture and human resources into a single dimension, corporate culture has a broader scope. It integrates not only human resources but also organizational processes, managerial procedures, physical infrastructure and technology (Gfrerer et al., 2021; Stoianova et al., 2020; Hoa & Tuyen, 2021). Based on this, the following critical factors for the success of DT in organizations are set forth:

- CF11. A corporate culture focused on operating in digital environments, promoting constant internal communication about innovations and business development (Fitzgerald et al., 2014; Mhlungu et al., 2019; Gfrerer et al., 2021; Stoianova et al., 2020; Wengler et al., 2021).
- CF12. A corporate culture that encourages employees to engage in innovation and experiment with emerging technologies and new business models (Mhlungu et al., 2019; Pramanik et al., 2019; Stoianova et al., 2020); Hoa & Tuyen, 2021).

Albeit there is a substantial amount of academic research dedicated to analyzing digital advancements, primarily within the domains of e-commerce and the retail sector (Kraus et al., 2021), research on the critical success factors of DT in organizations using the Delphi technique, particularly in retail, is an innovative approach that can contribute to the understanding of the topic and conceptualization of DT in organizational domains.

3. Methodological Procedure

The Delphi method is a practical approach based on the philosophical perspective of John Dewey, who believed that social science research should have a close relationship with the real-world practice and decision-making process (Brady, 2015). Delphi research is highly flexible and effective, and particularly useful in situations where there is insufficient knowledge about the phenomena to be studied (Skulmoski et al., 2007). It is also used to create future perspectives through a systematic construction of questioning and feedback reports guided by anonymous interactions among a group of carefully pre-selected experts (Lima et al., 2008).

Delphi studies have been used in the IS domain to improve the efficiency and effectiveness of technological and communications infrastructures in key sectors, as well as to enhance working relationships and facilitate group decision-making processes (Hartman & Baldwin, 1998; Brady, 2015). The method, which is extensively utilized in IS research, serves to pinpoint and prioritize critical issues in management studies (Delbecq et al., 1975; Addison, 2003). The method adopts an iterative approach to anonymously obtain expert opinions on a given topic via dispersed data collection and analysis techniques with feedback (Linstone & Turoff, 2002; Skinner et al., 2015). This procedure occurs in rounds and continues until a level of stability in the experts' responses is reached (Rowe & Wright, 1999). The Delphi method does not require expert panels to be statistically representative samples. Instead, the representativeness of the panel is assured through the experience and qualifications of the participating members and not the total number of same (Powell, 2003). According to Delbecq et al. (1975), once a homogeneous group exceeds 30 well-selected participants, there is little generation of new ideas for decision-making purposes.

To achieve its objective, the investigation aims to understand the importance of the dimensions involved in the DT process in organizations, by investigating the main drivers that promote successful DT initiatives. The primary distinction between the Delphi technique and other methods of collaborative decision-making is the absence of direct communication among the respondents (Delbecq et al., 1975). Conducting an in-person meeting opposes a fundamental principle of the Delphi procedure, which aims to prevent situations where a panel member might exert dominance over the consensus process (Boulkedid et al., 2011). Thus, an online questionnaire was formulated on the Qualtrics platform, by which experts were asked to evaluate, rank in order of significance and suggest the critical success factors associated with DT. The Delphi method suggests the inclusion in the sample of individuals with expertise, commonly referred to as "experts" (Skulmoski et al., 2007). By experts, it is considered retailers' executives and experts that own, run, work or provide consulting in companies that commercialize consumer goods to final consumers in physical and/or digital environment, mainly, but not exclusively, on the following segments: supermarkets, wholesales, drugstores, marketplaces, and convenience stores.

A direct invitation to participate in the survey was extended to around 300 retail industry experts, contacted mainly through accessibility sampling and retail groups on professional social networks, i.e. LinkedIn. Respondents were presented with 12 statements in the five dimensions described in the theoretical foundation section, aiming to assess their level of agreement on the critical factors previously outlined. To this end, a five-point Likert scale was used, ranging from "completely disagree" (lower limit) to "completely agree" (upper limit). Demographic questions, including education, gender, commerce segment, years of professional experience and company size were also included to characterize the profile of the participants in the research. Furthermore, respondents had the option to indicate if they found any questions unclear and were also encouraged to include new criteria that they considered critical. To ensure the validity of the questionnaire, it was distributed to a test group comprising five experts who proposed minor adjustments that were considered in the final version. The test group included academics well-versed in the Delphi method and experienced retail professionals.

3.1 Sample and Data Collection

In the first round, 75 respondents agreed to participate and responded to the full questionnaire. However, some respondents did not match the required profile and so their responses were discarded. The final number of subjects consisted of 52 respondents. This quantity suffices as the methodology hinges not on the absolute number of participants but rather on their capacity to articulate and offer feedback on a specific topic while remaining anonymous. Skulmoski et al. (2007) described 19 studies using Delphi with an average number of 23 subjects in each. Upon completing the initial phase of data collection, statistical methods were employed to conclude the initial ranking analysis. Four essential characteristics are typically required to define the Delphi method: anonymity, iteration, controlled feedback, and the statistical aggregation of group responses (Rowe & Wright, 1999).

Then, in a subsequent round, a follow-up questionnaire was dispensed to the same group of experts, being them presented to the results of the first round and asked to evaluate (re-rank) the previous rank obtained. They were also asked to propose any variable that could be considered a new critical success factor and that had not been considered in the first round. With that completed, 31 respondents agreed to participate in the second round. The results from each round were organized and examined to facilitate a comprehensive discussion of the outcomes derived from the two rounds of the Delphi method. The Delphi method seeks group consensus, so that the procedure (rounds) is completed when a certain level of agreement is reached among panel members (Boulkedid et al., 2011). To assess whether the rankings in the second round reflected a consensus among the group members, the Kendall's W-value, also known as Kendall's coefficient of concordance (Okoli & Pawlowski, 2004), was calculated. Subsequently, chi-square statistics were computed to obtain the *p*-value, allowing for an assessment of the significance of the W-value (Ameyaw et al., 2016; Yaokumah et al., 2023). The chi-square test for independence serves as a nonparametric tool to evaluate the presence of a relationship between two variables, and it has been used to examine the consistency of responses across the Delphi rounds (Dajani et al., 1979). Through this approach, it becomes feasible to evaluate the order of the importance as well as to set a comprehensive list of critical factors for DT in the organizational environment, as perceived by retail companies.

From the 52 respondents, all were very experienced professionals or consultants in the retail/commerce industry in Brazil, being 73% with 20 or more years of experience. Considering their gender, 23% are female, and 77% are male. Although there is a gap between genders, the result is compatible with the average proportion of 28% of women in higher positions in companies (Krivkovich et al., 2020). As for their job position, most of the total respondents are in high level positions: 56% are C-level, 27% are managers, 9% are from supervisors to analysts, and 8% are external

consultants, and 25% of all hold a master's or doctoral degree. Considering the segment of their businesses in the retail sector, 52% are supermarkets/wholesalers, 27% are drugstores, and 21% are others (including apparel, c-store, general merchandise, and tech). Organization size was also measured - 69% of respondents are in the largest companies (with more than 500 employees), 6% in companies with 100-499 employees and 25% in small and medium-sized companies.

3.2 Data Analysis

Data analysis was conducted following each round of the Delphi method, emphasizing the significance of intermediate statistical assessments (Franklin & Hart, 2007). Kendall's W statistical test was utilized to assess the degree of convergence among consecutive rankings provided by specialists. The use of the W statistical test facilitates the assessment of the consensus levels among participants, where W corresponds to 0.1 (very weak), 0.3 (weak), 0.5 (moderate), 0.7 (strong), and 0.9 (very strong) (Schmidt, 1997). Once a sufficient level of consensus is established between consecutive rounds, the iterations are halted. Below are the detailed outcomes obtained from the Delphi rounds.

4. Results

4.1 Delphi Round One

Table 1 displays the first assessment round of the 12 criteria arranged in descending order of their final averages.

Table 1: Ranked averages of the 12 criteria of Critical Success Factors for DT in Round One

Ranking	Critical Success Factors (CSF) for Digital Transformation in Retail/Commerce	Average (1 to 5)	Dimension
1	Leaders oriented towards promoting the implementation of new technologies and open to innovative ideas and actions.	4.94	Management Capability
2	Constant assessment of the company's digital evolution and identification of new opportunities that could become digital business models.	4.81	Business Processes
3	Employees with the ability to work with digital resources.	4.77	Human Resources and Capability
4	Leaders with knowledge of new technologies and their application to the business.	4.75	Management Capability
5	A corporate culture focused on operating in digital environments, promoting constant internal communication about innovations and business development.	4.73	Corporate Culture
6	Communication processes in digital businesses, such as acting on social media to create value and engagement with consumers.	4.69	Business Processes
7	Training and skills development programs for working with digital resources.	4.60	Human Resources and Capability
8	A corporate culture that encourages employees to engage in innovation and experiment with emerging technologies and new business models.	4.56	Corporate Culture
9	Technology infrastructure to support the business, such as Business Intelligence (BI) tools, Cloud, other digital automation, digitization, and management tools.	4.48	Technological Resources
10	Multiple sales channels (such as physical stores and online stores, different social networks, etc.).	4.48	Business Processes
11	Use of E-commerce platforms (e-commerce sites, sales apps, or other mobile businesses such as WhatsApp).	4.42	Technological Resources
12	Use of the latest innovative technologies such as Artificial Intelligence (AI), machine learning, augmented reality, robots, and others.	3.79	Technological Resources

In that round, some participants suggested the following new factors, which were then included to compose the ranking that was subsequently presented on the second round: (a) Customer-centric approach incorporating a vision of customer experience (CX/UX), with consistent key performance indicators (KPIs) for measuring their satisfaction; (b) Business-orientated technology, employing

digital models that promote business sustainability; (c) Fostering initiatives aimed at reducing challenges encountered by employees in learning and utilizing new technologies; and (d) Establishing partnerships to facilitate the implementation of high-investment technologies in small to medium-sized (SME) enterprises. The new factors were related to the following new dimensions, according to their main themes: (a) Customer Experience, (b) Business Sustainability, (c) Digital Inclusion, and (d) Financial Resources. They will be discussed next in section 5.

4.2 Delphi Round Two

The ensuing phase entailed soliciting participants to assess the final ranking accrued from the preceding round. Participants had access to anonymized comments provided by other specialists, with the aim of agreeing with the current ranking, suggesting potential reordering, or introducing new factors. The comparison between the first and the second ranking is delineated in Table 2.

Table 2: Ranking of the Critical Success Factors in Round One compared to Round Two

Dimension	Description of the Critical Success Factor for DT in Retail	Ranking Order Average	Ranking Order (Round 2)	Ranking Order (Round 1)	Variation
Management Capability	Leaders oriented towards promoting the implementation of new technologies and open to innovative ideas and actions.	1.48	1	1	0
Business Processes	Constant assessment of the company's digital evolution and identification of new opportunities that could become digital business models.	3.26	2	2	0
Corporate Culture	A corporate culture focused on operating in digital environments, promoting constant internal communication about innovations and business development.	4.29	3	5	2
Human Resources and Capability	Employees with the ability to work with digital resources.	4.81	4	3	-1
Management Capability	Leaders with knowledge of new technologies and their application to the business.	4.97	5	4	-1
Technological Resources	Technology infrastructure to support the business, such as Business Intelligence (BI) tools, Cloud, other digital automation, digitization, and management tools.	6.90	6	9	3
Business Processes	Communication processes in digital businesses, such as acting on social media to create value and engagement with consumers.	7.16	7	6	-1
Human Resources and Capability	Training and skills development programs for working with digital resources.	7.65	8	7	-1
Corporate Culture	A corporate culture that encourages employees to engage in innovation and experiment with emerging technologies and new business models.	8.87	9	8	-1
Technological Resources	Use of E-commerce platforms (e-commerce sites, sales apps, or other mobile businesses).	11.00	10	11	1
Business Processes	Multiple sales channels (such as physical stores and online stores, different social networks).	11.71	11	10	-1
Technological Resources	Use of the latest innovative technologies such as Artificial Intelligence (AI), machine learning, augmented reality, robots, and others.	12.71	12	12	0
Customer Experience	Customer-centric approach incorporating a vision of customer experience (CX/UX), with consistent key performance indicators (KPIs) for measuring their satisfaction.	12.81	13	13	0
Digital Sustainability	Business-orientated technology, employing digital models that promote business sustainability.	12.97	14	14	0
Digital Inclusion	Fostering initiatives aimed at reducing challenges encountered by employees in learning and utilizing new technologies.	14.58	15	15	0

Financial Resources	Establishing partnerships to facilitate the implementation of high-investment technologies in small to medium-sized (SME) enterprises.	15.23	16	16	0

Table 2 displays the average of the ranking order evaluated in the second round in column 3, and the correspondent rankings from the second and the first rounds in columns 4 and 5, respectively. Column 6 presents the variation between the outcomes of the initial and subsequent rounds. As expected in the Delphi method, it is necessary to assess the convergence of participants' opinions at each round following the initial one. Therefore, the statistical test of Kendall's W was applied at the end of the second round, presenting a value of 0.822, thereby indicating that the experts achieved a high level of agreement in this round. Schmidt (1997) proposes that a W value of 0.7 or higher is considered a strong agreement, with a high level of confidence in ranks. Additionally, the chi-square statistic was computed to evaluate the statistical significance of the W value (382.315), resulting in *p*-value of 0.000, suggesting a very high significance for the value of W. The values are consolidated in Table 3.

Table 3: Second round statistics and participants' level of agreement

Second Round Statistics	
Number of dimensions	16
Number of participants	31
Level of agreement (W)	0.822
Significance level (chi-square)	382.315
p-value	0.000

After verifying the level of agreement in the second round, Kendall's Tau (T) was used to verify the correlation coefficient between the consecutive rounds. The greater the T value, the stronger the consensus between the rounds, where a value of 1 corresponds to complete consensus. The T value obtained was 0.499, as described on Table 4. This way, according to the Delphi prerogatives, the high coefficient shows that it was not necessary to carry on another round.

Table 4. Correlation between the rankings

	Values
Number of dimensions	16
Kendall's coefficient of correlation (T)	0.499
p-value	0.001

5. Discussion and Conclusions

This study aims to offer an in-depth understanding of the critical success factors for digital transformation in organizations in the retail sector. The research was held in Brazil, a country that is considered an emerging economy, yet that embraces two worlds within one: highly developed and very poor societies coexisting. These disparities are also reflected in the retail/commerce businesses, from high-end stores and e-commerce platforms to very simple and informal street commerce. Moreover, in the 1990s, Brazil underwent significant economic reforms that liberalized markets. The country also reduced the wage inequality, unlike other Latin American countries, due to the large increase in the supply of more educated workers (Firpo & Portella, 2019). This growth diminished inequality through both price and composition effects, as explained by Barros et al. (2010). In this same period, Brazil

and other Latin American countries opened its retail sector to foreign direct investment with the expectation that the entry and expansion of retail chains would stimulate the development of a sector that had long been characterized by small family-run stores (Reardon et al., 2003; Vries, 2008). Moreover, some companies started to adopt new Information and Communication Technologies (ICTs). However, small independent retail stores were not replaced by international or larger chains, resulting in little productivity growth (Vries, 2008; Baer, 2001).

Since that period, Brazilian retailers have been trying to specialize themselves searching for the latest innovation in ICTs and business processes in developed countries in events like the Food Marketing Institute annual conferences held in the 2000s, and more recently the National Retail Federation (NRF) Retail's Big Show annual conference, both in the United States. The latest edition of NRF in 2024 registered almost 2,500 Brazilian participants, the largest international attendance (National Retail Federation, 2024). These diverse characteristics of the Brazilian retail segment demonstrate a valuable diverse and experienced market for the insights of the present study.

The results drawn from the Delphi method helped to unveil that the top positions are related to the Management Capability and Business Process dimensions. Moreover, the top five CSFs remained the same in both expert-led rounds with some minor position differences. Interestingly, Technology dimension is not in the top positions.

The first CSF refers to the management team's ability to promote and implement new digital technologies, as well as being open to innovative ideas and actions. Therefore, the management team's ability to remain receptive to innovative ideas and actions is critical to organizational success in DT initiatives. Scholars emphasize that the proactive involvement of the management team promotes technological adoption and innovation (Eisenhardt & Martin, 2000). Furthermore, the ability of organizational leaders to adopt new technologies and encourage innovation plays a central role in competitiveness in today's highly dynamic business environment. Thus, these results duly comply with the very fact that investing exclusively in ICT is insufficient for the success of DT initiatives, which requires a change in top management's mindset regarding the importance of emerging technologies. Although SMEs usually do not have well-defined and structured levels of management (Pirola et al., 2019), according to the perception of SME experts, management plays a crucial role in promoting DT.

The second most important CSF is related to the Business Process' dimension, addressing the constant assessment of the company's digital evolution and identification of new opportunities that can provide innovative digital business models. In fact, Soh et al. (2018) emphasize the importance of organizations monitoring their digital maturity levels and strategically aligning their digital initiatives with evolving market trends. Accordingly, digital innovation plays a fundamental role in the development of new business models, requiring companies to constantly monitor and implement new digital opportunities. An example of a retail company's constant search for digital evolution, in the search for new technological opportunities, can be seen in the discussion about whether the organization should become completely digital, maintain physical stores or adopt both business models, from an omnichannel perspective (Sousa et al., 2021; Teixeira et al., 2022). Although academic discussions on omnichannel retail are predominantly conducted from the perspective of developed nations - with the main studies being conducted in Europe and North America, with empirical data limited to emerging economies (Galipoglu et al, 2018; Sousa et al, 2021) - this result is in tandem with major international trends, as this topic was addressed in the discussion about the second most relevant CSF in retail in Brazil.

The subsequent positions are occupied by factors related to the dimensions Corporate Culture, Human Resources and Capability, and Management Capability. The third CSF highlights the importance of communication to foster a digital corporate culture, by informing about innovations and the related business, as part of a major digital environment. A corporate inclination for technology has a positive and significant impact on organizational digital culture (Bhatta et al., 2023) and as

companies acknowledge the necessity of excelling in a competitive landscape, they aim to develop sustainable digital culture, characterized not only by implementing and fostering innovation, but also by ensuring strong and consistent communication processes throughout the organization and stakeholders (Wengler et al., 2021). Communication is thus paramount in crafting and strengthening a sustainable corporate culture.

The fourth most important CSFs is embedded in the dimension Human Capabilities and Knowledge, namely the employees' ability to work with digital resources. This factor highlights the importance of organizations undergoing digital transformation initiatives to ensure the existence of qualified personnel capable of operating in new digital scenarios. This concern is specially challenging in the context of emerging countries, where the digital divide is broader, so people have limited access to ICTs and, consequently, less ability to utilize them proficiently (Madon et al., 2009). According to the Regional Center for Studies for the Development of the Information Society (Cetic.Br, 2020), the main reasons for citizens not to use the internet in Brazil are: the lack of Internet skills and the (high) cost of Internet, which undermines the development of digitally prepared workforce.

The fifth most important CSF is related to Management Capability, concerning leadership's knowledge about new technologies and how they can be applied to the business. Conversely, according to Mhlungu et al. (2019) technology awareness of the leaders is not a significant factor given that the leadership can rely on staff skilled in technology to advise and drive DT process. However, from the perspective of retail experts in this investigation, the perception was opposite, and technical knowledge on the part of leadership is seen as paramount, also playing a crucial role in mitigating potential strategic failures.

Another point of attention in this research is the lower position assigned to the multiple sales channels or omnichannel strategy. Although this is a widely debated topic among practitioners, this factor was only ranked 11$^{th.}$ position. In fact, more than a decade ago, Bharadwaj et al. (2013) observed that DT was reshaping market dynamics, leading to significant shifts in power across channels and disrupting traditional sources of economic profit.

Technology was long understood by companies as the leading driver if not the only for digital transformation until recent times. Works from Westerman et al. (2011) and Brynjolfsson and McAfee (2014) indicated that technological resources were the primary enabler for digital transformation within companies. In the same vein, albeit the study of DT readiness in SMEs proposed by Hoa & Tuyen (2021) list variables such as Leadership, Human Resources and Innovation Culture, they consider all dimensions and variables as related to the use of digital technologies. However, results show that the Technological Resources dimension only appears in the 6$^{th.}$, 10$^{th.}$ and 12$^{th.}$ positions in the rank. Albeit this corroborates with the sociomaterialist theories in IS studies— which refutes the idea of having technology as paramount above other actors (Hassan & Hovorka, 2011) — Technology far from the top positions is particularly interesting, indicating that the retail business in the context of an emerging economy may have reached a level of DT maturity comparable to that of its more developed counterparts. However, as described previously, the Brazilian reality simultaneously involves different contexts associated with developed and emerging markets.

In that sense, another aspect that draws attention is the lower importance assigned to successful use of most recent innovative technologies such as AI, machine learning, and augmented reality. Although several organizations have, at least, heard about these novelties, the use of them, according to the results, is still a distant reality. In developed economies, such as US, by incorporating AI and cameras, some retail chains like Amazon and Albertsons are exploring grab-and-go alternatives, raising thus the level of convenience of self-checkout (Terdiman, 2018). Furthermore, Kate Spade and IKEA are using augmented reality technologies to assist customers in their purchasing processes and other high-end technologies as narrowcasting enabled by AI has started to be discussed in the retail sector (Grewal et al., 2020).

However, the latest technological novelties are far from being implemented in a large scale in the Brazilian retail industry. A recent survey on the adoption of artificial intelligence (AI) technologies in the Brazilian retail sector found that 84% of companies have not yet incorporated these tools into their operations (Exame, 2023), which highlights the gap between the operation of retail organizations in developed economies and their counterparts in developing economies. On the other hand, in the context of emerging economies, a renewed interest in retail is to be expected, as this sector is a catalyst for the development not only of business, but also of social inclusion (Sousa et al., 2021).

Furthermore, an in-depth analysis comparing the final ranking, the related literature and the explanations provided by the experts about the scores given to each factor in the qualitative part of the survey is proposed below (Table 5). The factors placed in the 13th to 16th positions are the new ones suggested by the experts in the second round, being them highlighted in grey.

Table 5. CSF for DT, Theoretical References and Related Comments from Experts

Final Ranking	Critical Success Factors (CSF)	Theoretical References	Dimension	Comments from Retail Experts (explanations)
1	Leaders oriented towards promoting the implementation of new technologies and open to innovative ideas and actions.	Fitzgerald et al. (2014); Agrawal et al. (2019); Pirola et al. (2019); Stoianova et al. (2020); Hoa & Tuyen (2021); Gfrerer et al. (2021); Wengler et al. (2021)	Management Capability	*- The President or CEO must be the sponsor of the DT.* *- For DT to happen, senior management must be convinced that this is the way forward.*
2	Constant assessment of the company's digital evolution and identification of opportunities that could lead to new digital business models.	Fitzgerald et al. (2014); Bharadwaj et al. (2013); Mhlungu et al. (2019); Bican & Brem (2020)	Business Processes	*- Vision of business-oriented technology to generate business ventures rather than just improving image. Because retail has very low margins.*
3	A corporate culture focused on operating in digital environments, promoting constant internal communication about innovations and business development.	Mhlungu et al. (2019); Gfrerer et al. (2021); Stoianova et al. (2020); Wengler et al. (2021)	Corporate Culture	*- It's essential to respect the company's culture while devising a plan to integrate innovation initiatives.* *- Strong human resources work to spread a culture of innovation, so that employees change the switch from the "old traditional way" to the "new economy".*
4	Employees with the ability to work with digital resources.	Agrawal et al. (2019); Pirola et al. (2019); Gfrerer et al. (2021); Stoianova et al. (2020); Wengler et al. (2021)	Human Resources and Capability	*- Technology alone is not enough: you need qualified people.* *- It depends more on people and processes than just the tool.*
5	Leaders with knowledge of new technologies and their application in their businesses	Hansen & Sia (2015); Mhlungu et al. (2019); Stoianova et al. (2020); Hoa & Tuyen (2021); Gfrerer et al. (2021); Wengler et al. (2021)	Management Capability	*- Targeting with know-how reduces the possibility of error.*
6	Technological infrastructure to support the business, such as Business Intelligence (BI) tools, Cloud, other digital automation, digitization, and management tools.	Fitzgerald et al. (2014); Agrawal et al. (2019); Hoa & Tuyen (2019); Stoianova et al. (2020); Bican & Brem (2020); Gong & Ribiere (2021); Wengler et al. (2021)	Technological Resources	*- IT is the basis for DT, but people are essential.* *- Technology is fundamental, but doing simple tasks well can be the best innovation.*
7	Communication processes in digital businesses, such as acting on social media to create value and engagement with consumers.	Fitzgerald et al. (2014); Bharadwaj et al. (2013); Mhlungu et al. (2019); Pelletier & Cloutier (2019); Gfrerer et al. (2021)	Business Processes	*- DT needs planning, joint action in all communication channels and people to act with and support customers.*
8	Training and skills development programs for working with digital resources.	Pirola et al. (2019); Stoianova et al. (2020); Gfrerer et al. (2021)	Human Resources and Capability	*- You need a qualified and trained team so that the tool(s) can be used effectively and coherently.* *- You don't need training structure in-house, but it's essential that you do it, even if it's through a third party.*
9	A corporate culture that encourages employees to engage in innovation and experiment with emerging technologies and new business models.	Mhlungu et al. (2019); Pramanik et al. (2019); Stoianova et al. (2020); Hoa & Tuyen (2021)	Corporate Culture	*- To prepare the company to adapt to this new culture of constant innovation, many bureaucratic actions will have to be rethought in favour of the agility needed for the changes.*

10	Use of e-commerce platforms (e-commerce sites, sales apps, or other mobile businesses).	Fitzgerald et al. (2014); Hansen & Sia (2015)	Technological Resources	*No comments*
11	Multiple sales channels, such as physical stores, online stores, and social media.	Hansen & Sia (2015); Mhlungu et al. (2019)	Business Processes	*- Phygital is something that is in a constant development. A progress trend.* *- Bringing channels together is important. Online and offline must go hand in hand.*
12	Use of the latest innovative technologies such as Artificial Intelligence (AI), machine learning, augmented reality, robots, and others.	Fitzgerald et al. (2014); Agrawal et al. (2019); Pramanik et al. (2019); Hoa & Tuyen (2021); Ghobakhloo & Iranmanesh (2021)	Technological Resources	*- IA is a DT that is here to stay and innovate in many segments. It depends a lot on the size of the company.* *- These tools are an evolution, but I can make the transformation without them.*
13	Customer-centric approach incorporating a vision of customer experience (CX), with consistent key performance indicators for measuring their satisfaction.	Fitzgerald et al. (2014); Hartl & Hees (2017); Mhlungu et al. (2019); Pramanik et al. (2019); Hoa & Tuyen (2021)	Customer Experience	*-The CX vision is indispensable in the DT process too.* *- The CX must be a central concern in the process. Customers always at the center of decisions.*
14	Business-orientated technology, employing digital models that promote business sustainability.	Bican & Brem (2020); Hoa & Tuyen (2021); Feroz et al. (2023)	Business Sustainability	*- Using digital models for business sustainability.*
15	Fostering initiatives aimed at reducing challenges encountered by employees in learning and utilizing new technologies.	-	Digital Inclusion	*- We have a large number of employees over 50 who have difficulties with DT and need attention not to feel excluded, observing their difficulties.*
16	Establishing partnerships to facilitate the implementation of high-investment technologies in small to medium-sized enterprises (SME).	-	Financial Resources	*- Seeking partnerships to enable small/medium businesses to make investments in high-cost technologies.*

The articles referred to in Table 5 were identified as most related to TD in organizations, from the perspective of business processes. Factors 1 to 7 and 12 of the FCS were those that had the most solid discussion in the referenced literature, although none of them suggests an order of importance for the associated factors or dimensions, as proposed in the present study.

The vision of having the leadership sponsoring and conducting the process and being open to innovation above other factors had its importance being described in previous studies and has also stood out among organizations in an emerging context. Wengler et al. (2021) emphasizes that some of the most important success factors of DT may be influenced and are directly controlled by the management team. Fitzgerald et al. (2014) and Agrawal et al. (2019) corroborate the importance attributed to the top managing role, alerting for the risks of failure in DT processes as a result of lack of sense of urgency, digital vision and adequate leadership.

Moreover, the constant assessment of the company's digital evolution and identification of new business opportunities as the second most relevant CSF is in line with the need of a systematic approach to identify obstacles and create the necessary environment for the adoption of digital environment, as proposed by Agrawal et al. (2019). Albeit their work is more focused on the perspective of DT in supply chain, it provides valuable insights related to the business processes of assessing and identifying new digital opportunities. Experts also pointed the importance of business-oriented technology to generate business ventures rather than just improving the corporate image due to the challenges in operating in a highly competitive environment of very low margins. Bican & Brem (2020) highlight this issue, arguing that although companies with low margins are the most threatened by "digital", their transition to a digital business model can facilitate their repositioning in the market.

Corporate culture is a construct with multiple definitions, and the components related to it appear in different configurations in the organizational literature (Guiso et al., 2015). According to retail experts, having a corporate culture that is focused on operating in digital environments is

essential. This idea is ranked in the third position, and the experts have different approaches to it. Some suggest that it is crucial to respect the company's culture while devising a plan to integrate innovation initiatives ('It's essential to respect the company's culture while devising a plan to integrate innovation initiatives'). Others believe that strong human resources can work to spread a culture of innovation ('Strong human resources work to spread a culture of innovation'). The importance of corporate culture is also noted in the literature, particularly in the work of Stoianova et al. (2020), who designate it as the umbrella concept for embracing all other aspects that lead to DT in organizations.

Although the Technological Resources dimension is frequently mentioned in many references used, the CSFs associated with this dimension were only classified in 6th, 10th. and 12th. positions. According to the experts, although IT is fundamental to DT initiatives, companies could develop them without having to work with the latest innovations, such as AI. The 10th factor - use of e-commerce platforms (e-commerce sites, sales applications or other mobile businesses) - is the only one that did not raise any consideration from the experts. Thus, it can be realized that the use of digital sales platforms is no longer part of discussions about DT. In the literature under analysis, there are also limited references on this subject.

Finally, the experts suggested four new CSFs for successful DT initiatives. Although these factors may introduce new areas of exploration to strengthen the DT domain, it's important to note that only a few experts included them. Consequently, these new ideas were analyzed and related to novel dimensions that had not been accounted for in the prior assessment. This could be due to their absence in the reviewed academic literature or due to disparities in the approach to DT. The findings serve as a starting point for further exploration of these themes for academic and managerial purposes.

The first included factor (ranked 13th), "Customer-centric approach incorporating a vision of customer experience (CX), with consistent key performance indicators for measuring their satisfaction" was assigned to the dimension of Customer Experience (CX). CX has been a theme of increasing interest among academics and practitioners in the last years (Gentile et al., 2007; Lemon & Verhoef, 2016). Andajani (2015, p.630) conceptualizes CX as 'the stimulation done by the company against the senses of consumers.' Considering the retail context, Verhoef et al. (2009, p. 32) define CX as 'a multidimensional construct and specifically state that the customer experience construct is holistic in nature and involves the customer's cognitive, affective, emotional, social, and physical responses to the retailer.' Some experts considered the CX vision as indispensable in the DT process, being same a central concern in the DT process. However, CX as a critical factor for DT success is not an issue extensively exploited in the reviewed literature. In many cases, CX is not necessarily related to digital technologies, as the experiential resource often plays a more significant role, as noted by Gentile et al. (2007).

The second factor included (ranked 14th) refers to 'Business-orientated technology, employing digital models that promote business sustainability'. Being it categorized in the Business Sustainability dimension. Although organizational sustainability has emerged in the academic literature, the construct remains very remotely explored as an enabler for DT, as proposed by Bican & Brem (2020); Hoa & Tuyen (2021); and Feroz et al. (2023). Hoa & Tuyen (2021) have one of the few works addressing the themes ODT and Sustainability as well as the influence of DT readiness on businesses sustainable development. Nonetheless, as an unexpected outcome, it is noteworthy that this theme has started to grab attention of some organizations.

The third new factor (ranked 15th) addresses the concerns towards the difficulties faced by employees in learning and using new technologies in the DT process. The explanation provided focused on the challenges faced by employees over 50 years old in the retail segment, materialized in two strands – the learning difficulty itself and the exclusion resulting from digital skills. There are a growing number of initiatives to hire senior employees over the age of 50 to 60 in organizations including retail stores in Brazil (Exame, 2019). However, age stereotypes intensify the negative

relationship between age and digital fluency (Hampel & Kunze, 2023), which may hamper attempts to digitally include older people. Although Hartl & Hees (2017) provides minimal reference to the subject, it was not considered an academic source for a success factor in ODT, and no other specific references were found.

The last factor presented and evaluated by the experts (ranked 16th) concerns the Financial Resources dimension, related to the importance of establishing partnerships to facilitate the implementation of high-investment technologies, mainly in SMEs. To date, no literature on ODT discussing this subject was found. Similar to the lack of references in the 15th. factor, the only work that refers to financing models and partnerships underlying successful ODT initiatives is provided by Mhlungu et al., (2019), which, however, does not relate with the specific needs of SMEs according to experts.

Finally, to structure the results obtained, the dimensions relevant to organizational DT were organized according to the average order of importance attributed by experts, being calculated from the scores of each CSF (Figure 2). Thus, Management Capability appears as the main dimension, followed by Human Resources, and Corporate Culture. The remaining four dimensions are visibly distant from those considered indispensable for the success of organizational TD, for the reasons mentioned above.

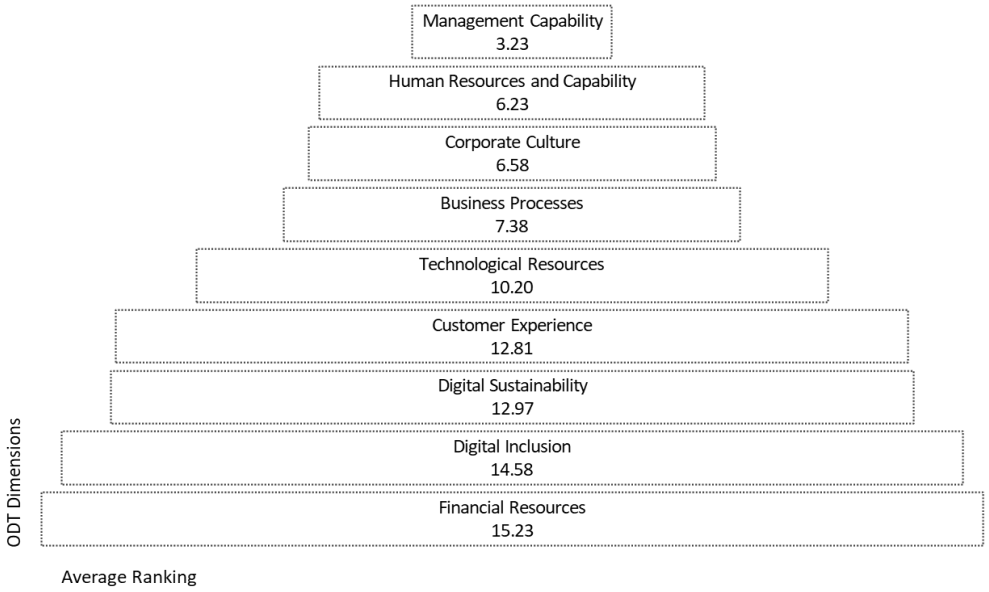


Figure 2. Pyramid of Organizational Digital Transformation Dimensions' Ranking

5.1 Theoretical and Practical Contributions

Academic research on the determinants of successful DT in organizations, particularly from a business standpoint, remains relatively sparse (Matt et al., 2015). This scarcity is evident in previous studies, such as the literature review on DT readiness in organizations carried out by Michelotto (2023). Furthermore, in a search on the Web of Science platform, only ten studies were found that, via the Delphi method, investigated DT in organizations. However, none addressed commercial companies, especially retailers (Web of Science, 2024). Thus, by delving into the CSF for DT, this research seeks to contribute to the existing literature on the subject by improving the integration of complex and diverse fields of knowledge related to DT in organizations, namely Management and Leadership, Business Processes, Corporate Culture, and Human Resources. Therefore, this study aims

to understand how companies rank the importance of dimensions related to organizational DT readiness. This work also brings a new configuration of factors - and their respective dimensions - that are considered paramount for the success of DT initiatives in companies. It also aims to encourage debate not only about what they are, their descriptions and conceptualization, but also to broaden the understanding of how they are being managed or, at least, whether or not they are being managed accordingly by practitioners.

Also, this paper provides intriguing insights for managers and entrepreneurs seeking to deepen their understanding of DT in organizations. From a managerial perspective, this work sets forth that human-related factors play a central role amid the multiple challenges that retail organizations face in their DT process, being same a key distinguishing factor in the success of DT initiatives. Indeed, organizations in general, rather than just retailers, can benefit from this study, as it identifies and classifies the key dimensions for organizational strategic decisions about DT in today's digital society.

The joint analysis of the dimensions related to digital transformation processes is a great instrument to be used by companies at any stage, whether they configure large or small organizations. In this sense, not only large corporations would benefit from the findings, but also SMEs that are taking initial steps in the so-called fourth Industrial Revolution (Pirola et al., 2019). The results of the survey with retail experts validate the importance of focused and well-prepared leadership, proper business processes, skilled employees, and organizational culture grounded on a digital mindset. These prerogatives are fundamental to boost investments that allow the design of business models capable of operating in competitive digital markets, such as investments in digital training and R&D. Furthermore, although technological resources constitute a fundamental and indispensable part of these processes, human and cultural factors have prevailed when thinking about DT and innovation.

Finally, the organized ranking with its related dimensions can be also useful to help companies in their strategic planning and execution, not only to transform existing business structures but also create complete new digital business models via the implementation of the latest technological innovation. Moreover, the understanding of the CSF associated with DT can promote innovation by unveiling opportunities to implement emerging digital technologies. Furthermore, it can facilitate knowledge sharing and capability development initiatives, encouraging companies to compete more effectively in a globalized digital economy.

5.2 Research Limitations and Further Steps

As any research within the social sciences field, this study is subject to certain limitations. The Delphi method, while widely used for assessing Critical Success Factors (CSFs), has its limitations. The number of participants from the first to the second round diminished, as a difficulty to access professionals/experts to answer surveys, notwithstanding the final number in this research has reached the acceptable amount for credible analysis. Moreover, reliance on expert opinion may overlook diverse perspectives or emerging trends. Nonetheless, the Delphi method is considered to be valuable to develop consensus-building and forecasting (Rowe & Wright, 1999).

Furthermore, the study did not examine potential variations in respondents' perspectives, concerning the type and size of companies in the retail sector. Future research may delve into this topic to elucidate potential distinctions that were not considered in the present study.

Besides, the absence of research on Latin American countries highlights the lack of specific attention given to this region. In the case of retail companies, given the economic significance of the region, this is somehow unexpected (Fastoso & Whitelock, 2011). In fact, Sousa et al. (2021) have

emphasized the need for further investigation in countries such as Brazil, which, besides being an emerging market, configures as top priority for retail expansion. This way, future studies in IS, including digital transformation and innovation, and organizational aspects such as in retail industry shall be expanded in the unique context of emerging countries. A more comprehensive investigation into the challenges of technological learning observed in countries with low educational and technological indicators could also justify further academic exploration.

As reviewed in the literature, authors have proposed different ways of assembling the constructs related to DT in organizations. Yet so far there is no unified definition for DT. In this sense, a more in-depth investigation into some polysemic constructs identified as critical factors for ODT, such as Digital Sustainability and Customer Experience, is also recommended.

Finally, although this work presents the organizational dimensions associated with DT, new studies may also deepen the understanding of how these dimensions are associated or merged, generating competitive advantage in an era characterized by constant changes and profound transformations. In sum, the ongoing journey through new organizational dynamics impacted by digital advances seems to be vast and long lasting.

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7. Paper III

DIGITAL TRANSFORMATION IN PAYMENT METHODS: THE SOCIAL REPRESENTATION OF PIX IN THE RETAIL INDUSTRY

Abstract

Organizations worldwide, in recent decades, have battled with the imperative challenge of digital transformation, being the financial sector no exception. The advent of technology-driven innovations in financial services suggests a path to promote inclusion, as many individuals lack access to them, especially in emerging economies. This is the proposal of Pix, the successful digital peer-to-peer instant payment system launched by the Brazilian Central Bank in 2020. However, while Pix has become a significant subject, there remains a research gap in understanding the perceptions surrounding it, not only among end-users but also from a business perspective. Thus, this article aims to contribute to the existing literature by Pix's social representation from an organizational point of view. To achieve this, one drew upon the Social Representation Theory (SRT), which is vastly used to explain how society perceives new digital innovations. Accordingly, this study relies on the diverse and prominent Brazilian Retail industry to deepen the understanding of how that new payment model is perceived by the market. Results reveal that attributes related to speed, convenience, and ease of use compose the central nucleus of Pix's social representation.

Keywords: Digital Payment, Peer-to-Peer Mobile Payment, Pix, Digital Transformation, Social Representation, Retail.

1. Introduction

In recent decades, organizations have faced the prevalent challenge of becoming digital (Henriette et al., 2016). Based on robust IT architecture, digital platforms allow users to access and deliver diverse and interconnected functionalities (Staykova & Damsgaard, 2021), being same not different in the financial domain. In fact, the emergence of technology-enabled innovations in financial services indicates that it is possible to align the inclusive direction of innovation with economic progress despite an estimated 1.4 billion people remaining unbanked and lacking access to affordable financial services (de Sant'Anna & Figueiredo). Nevertheless, when a new technology enters people's daily lives, it often triggers a mix of emotions among the public. While these innovations are designed to simplify everyday tasks and enhance comfort, they also evoke uncertainty in some people (Wagner & Kronberger, 2001). This has not been different with Pix, the instant peer-to-peer digital mobile payment system introduced by the Brazilian Central Bank in November 2020.

The launch of Pix enabled Brazil to achieve a 24/7 payment system that is free for users and benefits citizens and businesses as well (Lobo & Brandt, 2021; Mantovani & Lucas Jr, 2022). Considering its widespread public support and adoption, Pix has become a significantly relevant subject since its inception, leading the retail sector to embrace it. In fact, the Brazilian retail scenario has been defined by the expansion of physical stores and e-commerce, as well as technological

advances and digital inclusion policies that ended up increasing competition (Sousa et al., 2021). Consequently, the rapid transformations in the digital environment, including new financial payment models and how people and companies have been dealing with them, has required continuing studies to keep pace with same. Accordingly, there is a research gap in understanding the perception of this new digital payment method, Pix, not only in terms of its diffusion and acceptance by end users, but also from the perspective of businesses, considering the Brazilian context of emerging market.

The launch of Pix is aligned with one of the primary criteria for the emergence of a social representation (Wagner & Kronberger, 2001): understanding new phenomena in society. Social representations comprise diverse meanings and interpretations as they serve as categories for classifying phenomena as well as theories for establishing facts embedded in collective social reality (Jodelet & Moscovici, 1989). Accordingly, social representation offers a valuable framework for understanding how individuals perceive new developments, such as technological advancements and scientific breakthroughs (Kalampalikis et al., 2013). Thus, to contribute to the extant literature on the subject, this article aims to investigate the social representation of Pix using the Social Representation Theory developed by Moscovici (1961) and the Central Nucleus Theory proposed by Abric (1993), as the emergence of a novel situation or an unusual event is a focal point in social representation investigations (Moscovici, 1981).

The retail sector was chosen due to its significant socioeconomic impact and the remarkable changes in digital payments that it has undergone in the past few years. Brazil is an emerging economy with substantial socioeconomic disparities, also reflected in the retail/commerce businesses. In the 1990s, Brazil opened its retail sector to foreign direct investment, hoping that the entry and expansion of retail chains would stimulate the growth of an industry that had long been characterized by small family-run stores (Reardon et al., 2003; Vries, 2008). Besides, e-commerce started in Brazil in 1996, and one of the factors contributing to its evolution has been the emergence of Pix for online purchases.

At the beginning of 2021, the use of Pix in e-commerce was only 1% whereas in 2023 it had reached 26.2% (IstoéDinheiro, 2024). In the overall retail market, including physical stores, approximately 35% of all payment transactions in the 1st. quarter of 2023 were made via Pix (Supervarejo, 2023), an extremely significant value for retail payments. Moreover, this industry was also the focus of previous research, namely the second doctoral dissertation paper, which addressed the critical success factors for organizational digital transformation.

The present study follows the subsequent structure: section 2 introduces the theoretical background, section 3 describes the methodology adopted, section 4 presents the results, and section 5 discusses same and presents the conclusions, including a subsection with the limitations of the research, its main contributions for theory and practice, as well as a research agenda on that subject.

2. Theoretical Background

2.1 Social Representation Theory and its Relationship with ICT and IS Studies

Social Representation Theory (SRT) explores how people in a particular social context collectively construct and communicate shared meanings about various concepts, objects, or events (Moscovici, 1961). It is a framework rooted in social psychology that helps understand psycho-social phenomena within contemporary societies (Wagner & Kronberger, 2001), being this approach used as the main guide for the present study.

According to SRT, individuals construct mental representations of the world through social interactions, which shape their understanding and interpretation of reality. Thus, primary functions of social representations are "ways of making the world" (Moscovici, 1988, p. 231). Therefore, representations are generated by familiarizing with new elements of the physical and social world (Moscovici, 1961; 2000), from anchoring these elements to preexisting categories of common sense.

This procedure renders something unfamiliar into something known, enabling individuals to categorize and assign labels to the new entity based on established and mutually understood categories of ideas and representations (Moscovici, 1984). In this sense, Moscovici defines social representation as 'the elaborating of a social object by the community to behave and communicate' (Moscovici, 1963, p. 251).

The concept of social representations connects to Durkheim's idea of "collective representations," which denotes shared frameworks for comprehending, perceiving, and evaluating social reality (Durkheim, 1898). However, Moscovici (1989) proposes that Durkheim's concept appears somewhat static and failed to fully encompass the complexities of modern society at time. Thus, social representations involve individuals' perceptions of a specific social phenomenon, enabling an understanding of how social identity is shaped, transcending the simplistic notion of mirroring or duplicating an external reality. Instead, representations manifest a symbolic space wherein their development and negotiation occur (Farr, 1993; Voelklein & Howarth, 2005). The notion of a reciprocal enrichment of common sense and scientific knowledge intrigued Moscovici (Marková, 2017), and one of his major contributions to social representation studies was to recognize the uniqueness of Guttman's work that aimed to uncover the structures of items that unite respondents (Moscovici, 1961; Guttman, 1954; 1959).

Another notable contribution from Moscovici is the concept of cognitive polyphasia, which underscores the recognition of social knowledge's multifaceted and diverse character (Jovchelovitch, 2004). Moscovici posits that different and opposite cognitive styles and knowledge forms can coexist within a social group, even within a single individual (Jovchelovitch, 2002). Consequently, individuals can access and employ conflicting representations contingent upon their specific tasks and prevailing social contexts (Moscovici, 1961; Jovchelovitch, 2002; Voelklein & Howarth, 2005). Nevertheless, despite acknowledging the plurality and complexity inherent to social representations, they are susceptible to modification challenges and changes, as any idea formulated by human agency (Voelklein & Howarth, 2005).

Initially, the literature on social representation focused on human beings and their interactions within society, given the origins of SRT in psychoanalysis. However, over time, social representation has been increasingly applied to non-human entities (Joia & Michelotto, 2020). Jodelet & Moscovici (1989) advocate for this extension, arguing that a social representation encompasses the cognitive process through which a subject engages with an object. According to Jodelet (2001), social representations are based on understanding the construction and use of knowledge in everyday life, offering a theoretical framework for understanding individual behavior and historically contextualized social facts. In the context of organizational studies, for instance, SRT has been used to understand how employees perceive and interpret organizational changes, as well as how consumers form representations of brands and products (Kalampalikis et al., 2013).

Given its nature as an academic field of constant innovation, studies on Information and Communication Technology (ICT) and Information Systems (IS) have been persistently challenged to define emerging constructs arising from the appearance of new technologies. Therefore, the perspective of social representation provides an invaluable framework for comprehending how individuals perceive novel occurrences, such as the ones involving technological innovation (Kalampalikis et al., 2013). Indeed, a list of almost twenty works in the field of ICT/IS using social representation reinforces this rich association, as detailed in chronological order, in Table 1.

Table 1. Social Representation studies on ICT/IS

ICT/IS Themes	Authors
Biotechnologies	Wagner et al. (1996)
Internet	Capozza et al. (2003)
Work Practices with IT Use	Vaast & Walsham (2005)
Mobile Phones	Contarello et al. (2007)
Information Systems Implementation	Gal & Berente (2008)
Telecommunications	Fortunati & Manganelli (2008)
Social Media	Kaganer & Vaast (2010)
Virtual Community Participation	Bailey & Ngwenyama (2011)
Community Multimedia Centers	Rega et al. (2013)
Cybersecurity	Pawlowski & Jung (2015)
Citizen e-Participation	Pozzebon et al. (2016)
CIO Competencies	Joia & Correia (2018)
ICTs for Community Empowerment	Bacallao-Pino (2018)
Success in IT Projects	Joia & Melon (2020)
Cloud Computing	Joia & Marchisotti (2020)
Fintechs	Joia & Proença (2022)
Nanotechnologies	Brondi et al. (2022)
Smart Cities	Michelotto & Joia (2023)

2.2 Shaping Social Representations: Central Nucleus Theory

This Central Nucleus Theory is used to shape social representations. Developed by Abric (1993), the theory proposes a structural approach where some aspects of the representation, described as central, have the essential characteristic of generating the meaning of all (or a large proportion of) the elements of the representational field (Abric, 1996; Flament, 1987). According to this theory, the structure of a social representation comprises two internal systems, namely a central core (or nucleus) and a peripheral system (Abric, 2001). From a descriptive point of view, social representations are built upon knowledge and schemas generally presented in the form of naive, non-formalized notions: knowledge, images, attitudes, or beliefs (Guimelli & Rouquette, 1992).

The central nucleus has the function of providing significance to the social representation. It determines the consensus-engaging themes, notably influencing individuals' decision-making processes and actions (Abric, 2001; Wachelke, 2008). The central nucleus represents the immutable essence of a social representation, being stable and resistant to changes, which ensures stability to the social representation (Guimelli, 1993; Mazzotti, 2002). Additionally, it orientates the organization of the peripheral system, which surrounds the central nucleus and accommodates the variability and diversity within the social sphere (Lin et al., 2013).

In contrast, the peripheral system exhibits less stability than the central nucleus, functioning as an individual moderator to safeguard the integrity of the central nucleus's meaning (Moliner & Tafani, 1997). Besides, the peripheral system unites elements of daily life, without which the central nucleus cannot be understood (Flament, 1987; Mazzotti, 2002). Besides, the contrast zone and the first and second peripheries provide room for a less straightforward interpretation of social representation, being able to embody cognitions that diverge from the core (Abric, 2003). The foundation of the of

these four quadrants - central nucleus, contrast zone, and first and second peripheries - lies in a collection of cognitive elements identified through the free evocation technique. By means of that, participants are presented with a stimulus word and are instructed to promptly respond with the first words that come to their minds (Bodet et al., 2009; Vergara, 2005; Joia & Michelotto, 2020).

Additionally, from an epistemological perspective, the structural approach is a significant milestone for the SRT. Firstly, it offers a conceptual framework for studying stable representations rather than those in the formation process. Secondly, it provides an analytical framework for examining the interaction between individual acting and the social context in which individuals operate (Rateau & Lo Monaco, 2013).

2.3 Digital Payment Methods

Digital Payment and Mobile Payment

The worldwide dissemination and use of the Internet and mobile devices have significantly fostered digital transformation, including new financial transactions and payments methods (Alkhowaiter, 2020). Accordingly, ICT has become essential to 21st. century society, revolutionizing the banking and finance industry. In fact, financial institutions have provided various customer services based on innovative ICT, with digital payment solutions at the top (Kajol et al., 2022). Indeed, one of the expected features for citizens to be considered financially included is their ability to make electronic payments (de Sant'Anna & Figueiredo, 2024).

Digital payments can be described as 'any type of payment using digital instruments, which include mobile payment, mobile wallets, cryptocurrency, and electronic payment' (Alkhowaiter, 2020, p.1), while mobile payment (M-payment) is a payment service based on mobile devices, mainly smartphones, to offer a convenient and versatile solution for making payments for goods, services and bills (Kim et al., 2010). Mobile payment, thus, is considered a subcategory of digital payments (Alkhowaiter, 2020). Yet, despite its advantages and increasing prevalence among consumers as an emerging service, mobile payments were still distant from worldwide adoption some years ago (Patil et al., 2020). However, in recent years, digital payments have revolutionized the financial system of developing economies, providing 865 million people access to their first bank account. This was particularly true after the COVID-19 pandemic when the circulation of physical money was associated with a high risk of contamination (Demirgüç-Kunt et al., 2022; de Sant'Anna & Figueiredo, 2024).

Searching the Web of Science database (2024) through titles and abstracts, it becomes clear that academic literature has highlighted two primary constructs related to the use of payment methods based on digital technology — digital payment and mobile payment (or m-payment). The search for the digital payment construct brought 397 papers, while the mobile payment (or m-payment) construct brought 1,700 papers. From the total of 2,097 papers, 44 brought both constructs together. Then, a textual analysis was provided to identify the most cited constructs within those papers, by screening the abstracts of both queries. The abstracts accrued from digital payment papers counted 93,497 words, and the ones from mobile payment (or m-payment) papers counted 183,649 words.

The constructs that were presented in both singular/plural forms were consolidated into just one of the ways and were then ranked. The 25 primary constructs most cited in each group of papers are shown in Table 2, excluding the word 'payment,' which is the core word for both.

Table 2. Most cited constructs in papers related to digital payment and mobile payment

	Digital Payment		Mobile Payment
1	Digital	1	Mobile
2	Financial	2	Services
3	Technology	3	Users
4	System	4	Technology
5	Services	5	Intention
6	Adoption	6	System
7	Transactions	7	Adoption
8	Mobile	8	Consumer
9	Consumer	9	Security
10	Covid-19	10	Usage
11	Economy	11	Trust
12	Inclusion	12	Influence
13	India	13	M-payment
14	Customer	14	New
15	Social	15	Financial
16	Users	16	Risk
17	New	17	Transactions
18	Online	18	Social
19	Value	19	Acceptance
20	Money	20	Value
21	Intention	21	Online
22	Fintech	22	Information
23	Security	23	Digital
24	Banking	24	Privacy
25	Information	25	Structural

From the lists of the most cited words, 17 of them appear in both rankings: Adoption, Consumer, Digital, Financial, Information, Intention, Mobile, New, Online, Security, Services, Social, System, Technology, Transactions, Users, and Value. Several articles surveyed are based on the TAM and UTAUT theories. Another aspect of this review is the focus of papers on consumers in both lists. Several papers relate to the digitalization of payments during the COVID-19 pandemic, enabled by physical money circulation restrictions. Besides, issues related to transaction security and trust in digital payment/mobile methods, including privacy concerns, are also highlighted. Furthermore, the importance of fintechs in the digital transformation of payment methods, as well as in the digital inclusion of unbanked populations, are also at the top of the list.

New Brazilian Digital Payment: Pix

The monetary system, a crucial foundation for the economy, is supported as a public good by central banks that supervise the payment system, promoting its safety, integrity, and efficiency. In many economies, central banks go further by providing and operating essential payment infrastructure, committed to fostering an open payment platform that can serve users effectively (Duarte et al., 2022). The Brazilian regulatory framework established by Law No. 12,865/2013 has empowered the Central Bank of Brazil (BCB) to regulate retail payments, emphasizing efficiency, security, interoperability, and financial inclusion (Duarte et al., 2022). In fact, there has been a notable surge in technology integration within banking and payment services in recent years. However, it was only with the launch of Pix by

BCB in November 2020 that Brazil achieved a 24/7 payment system with real-time payment finality, benefitting both citizens and businesses (Lobo & Brandt, 2021).

Pix is a payment method created by the BCB, transferring funds between accounts in a few seconds at any time or day. On the Central Bank's official communication website, the main features related to Pix are described as 'practical, fast, and secure' (BCB, 2024, p.1). Pix can be linked to a current account, savings account, or prepaid payment account, and its goals include improving effectiveness and competitiveness, fostering the adoption of digital payment solutions, advancing financial inclusion, and filling the gaps in existing payment methods (BCB, 2024; Duarte et al., 2022). Additionally, Pix stands out from other payment and transfer methods in the country as it offers the flexibility of transferring funds and making payments on any day, including weekends and public holidays, and at any hour of the day. These features have differentiated Pix from other payment and transfer methods since its official launch (Rimonato & dos Santos, 2021).

Just one month after the system's launch, Pix reached 40 million users, and until March 2024, it accounted for a total of 162 million users, while the entire Brazilian population accounts for 203 million people (IBGE, 2022). From these, 71 million were new inclusions to the financial system (BCB, 2024). The success of Pix can be attributed to various factors, but two stand out— the mandatory participation of larger banks and institutions and the adequate governance arrangements that have been put in place to set the rule (Duarte et al., 2022). Moreover, it offers a cost reduction associated with intermediaries in the payment process, thereby providing a new competitive advantage for the financial sector (Schapiro et al., 2023). Another essential characteristic of the broad acceptance of Pix is related to its user-centric focus, which aims to meet the specific payment needs of households and businesses and fill the gaps in the current system (Duarte et al., 2022).

Surprisingly, the percentage of transactions in Brazil using Pix increased from 0.5% in 2020 to 40.9% in 2023. The evolution of the percentage of main payment methods and their total amount of transactions (in billions) in Brazil are detailed in Figure 1. According to the Brazilian Banking Association, Pix has been consolidated as the most widely used means of payment in the country (Febraban, 2024).

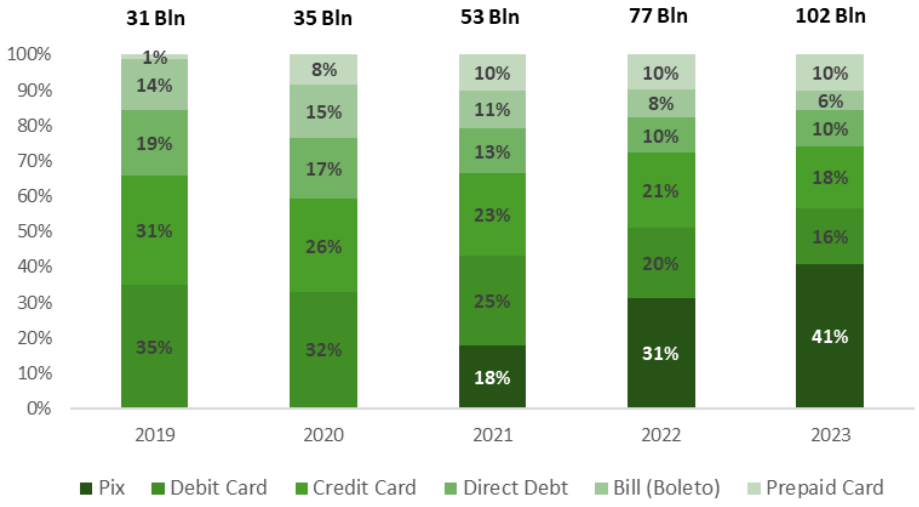


Figure 1. Evolution of the transactions of main payment methods in Brazil. Adapted from BCB (2024).

Pix's practicality, full-time availability, and free-of-charge to individuals have made it a convenient choice for consumers (Argentieri Mariani et al., 2023). To date, to make transactions, citizens have three access options: by bank details, Pix Key, or QR Code. According to the National Confederation of Retailers (CNDL), the QR Code used by the Central Bank - the so-called BR Code

- is a specific, open, and accessible standard that meets the requirements of the Brazilian payment system, making it possible to store multiple arrangements in a single code (CNDL, 2020). The Council of the Americas awarded BCB the Bravo Beacon of Innovation prize for Pix, recognized as a worldwide model of financial inclusion (Mercado e Consumo, 2024). In 2019, 77% of retail transactions in Brazil were in cash, 12% in debit cards, and 7% in credit cards (Lobo & Brandt, 2021). Debit and credit card transaction fees can be as high as 5% in retail stores. In e-commerce, instant payment is already 100% accepted, tied with credit cards (Mercado e Consumo, 2024). Since the beginning of e-commerce in Brazil, in 1996, the emergence of Pix has been the biggest innovation in terms of digital payments. Indeed, at the beginning of 2021, Pix represented only 1% of the payments in e-commerce; however, by 2023, it had reached 26.2% (IstoéDinheiro, 2024). In the overall retail market, including physical stores, approximately 35% of all payment transactions in the 1st. quarter of 2023 were made via Pix (Supervarejo, 2023), which is extremely significant in retail payments.

2.4 The Retail Industry in Brazil

The retail industry wields substantial global economic power and occupies a unique position between supply and demand, making it a key player in the market. Retailers can influence production and consumption patterns worldwide (Delai & Takahashi, 2013). Due to the rapid advancements in technology affecting the retail industry, experts and professionals in retailing strive to comprehensively understand the scope and nature of these effects through a systematic approach (Shankar et al., 2021). About twenty years ago, most attempts to transform brick-and-mortar stores into profitable online retailing failed, especially in grocery stores (Enders & Jelassi, 2009). Although many businesses have begun to adopt e-commerce, social media marketing, and analytics-based supply-chain management, only a few have successfully developed the necessary capabilities to meet the current demands (Begley et al., 2020).

Retail activity began in Brazil when the Portuguese royal family arrived in former colonial Brazil in 1808. The opening of ports caused an intense movement of ships and goods in Brazilian port cities, and European merchants came to settle in the central Brazilian cities, opening many shops in the streets close to the ports (Varotto, 2018). Since then, Brazilian retail has followed European and North American trends, adapting same to local reality. In the Brazilian context, several prominent and established companies can be observed today, and the retail scenario drastically varies from very simple small groceries to high-end superstores. The French-based Carrefour Group leads the ranking, responding to annual revenue of US$15.3 billion in 2022 (IBEVAR, 2022) in a highly concentrated market, with its leading players detailed in Table 3. As per the retail segment, the Supermarket/ Grocery segment leads the percentage of consumer goods consumption, followed by the Automotive segment, representing more than 60% of the retail consumption in Brazil (IBEVAR, 2022), as detailed in Table 4.

Table 3. Largest Retailers in Brazil. Adapted from IBEVAR FIA (2022).

	Segment	Revenue 2022 in US$ billion**
Carrefour Group	Supermarket/ Grocery	15.3
Assai	Supermarket/ Grocery	8.6
Magazine Luiza	Household Appliances	8.1
Via Varejo	Household Appliances	6.8
Lojas Americanas/B2W	Department Store	6.1
GPA (Pao de Açucar)	Supermarket/ Grocery	5.5
Raia Drogasil	Drugstore	4.8

Source IBEVAR FIA 2022 ** Currency 1 R$ Real = US$5.3

Table 4. Consumer Goods Consumption in Brazil (%). Adapted from IBEVAR FIA (2022).

	% Consumer Goods Consumption
Supermarket/ Grocery	36.2
Automotive Vehicles	26.9
Building Supply Stores/ DIY	8.6
Apparel	7.5
Drugstores	6.5
Fuel/ Gas Stations	5.3
Household Appliances/ Furniture	5.3

Source IBEVAR FIA 2022

The retail Supermarket/Grocery industry in Brazil contributes to 7% of the national GDP (of US$1.6 trillion) and provides income to 3.1 million individuals who directly or indirectly participate in an ecosystem that serves 28 million consumers per day, according to the 2022 ranking from the Brazilian Supermarket Association (ABRAS, 2023; World Bank Group, 2023). Technological changes, such as the emergence of the Internet, digital inclusion policies, the proliferation of physical and virtual sales channels, and intensified competition among retail companies, have characterized the Brazilian retail backdrop (Sousa et al., 2021). Besides, sophisticated information and management technologies have been adopted by the most prominent Brazilian retail companies to sustain their market positions and expand their operations (Sousa et al., 2021).

The Brazilian market represents a promising growth market for e-commerce operations (Sousa et al., 2021; JPMorgan, 2021). Therefore, Brazil is the only country in Latin America that ranks among the ten largest e-commerce markets in the world (Botelho & Guissoni, 2016). Indeed, during the first quarter of 2020, at the beginning of the pandemic, Brazilian e-commerce retailers, such as Magazine Luiza, B2W and Via Varejo, showed an increase in online sales of, respectively, 72.6%; 35% and 52% (Almeida et al., 2020).

The retail sector was chosen in this research due to its significant socioeconomic impact and remarkable shifts resulting from digital transformation initiatives, including changes to digital payments that the sector has undergone in recent years. In fact, Brazil is an emerging economy with substantial socioeconomic disparities, which is also reflected in the retail businesses. However, due to its importance, the dynamic Brazilian retail sector is a highly relevant market for studies related to the evolution of digital payments.

3. Methodological Procedures

The investigation employed a mixed-methods approach, applying both qualitative and quantitative methods to determine the social representation of Pix and the perceptions of the respondents related to this new digital payment method. The first step consisted of conducting a survey divided into three parts: (1) the free word evocation technique and the assessment of the main reasons for the evoked words, (2) open questions about the understanding of Pix in the organizational process of DT and as a factor for digital inclusion and (3) socio-demographic data related to the participants.

Word Evocation Technique and Vergès' Quadrant

Given that social representation stems from a stimulus word, one of the techniques employed is the word evocation technique, which involves gathering words/expressions articulated by respondents when presented with a specific suggestive word or expression (Vergara, 2005; Joia & Michelotto, 2020). Usually, five words/expressions are asked, written down one after the other (Gaymard & Joly, 2012). Thus, the following question was applied: "*When you think of Pix, what are the first five words or expressions that immediately come to your mind?*"

To organize the social representation, after listing all the participants' evocations, Vergès (1992) proposes the notion of prototypicality, which requires the combination of two analyses—a lexicographical analysis (frequency of occurrence of terms) and the analysis of the order of appearance of the terms evoked. This approach identifies four distinct zones - Central Nucleus, First Periphery, Contrast Zone, and Second Periphery (or Peripheral System). Then, once the words are analyzed, it is possible to define the central nucleus and the peripheral system to outline the graphic scheme of the Vergès' Quadrant. The Central Nucleus encompasses the most frequently cited terms, typically in the first ranks. The First Periphery comprises terms that are cited frequently but are located in the last ranks or cited infrequently and found in the first ranks. The Contrast Zone includes infrequent but essential elements, and the Second Periphery encompasses infrequent terms in the lowest ranks (Gaymard & Joly, 2012; Brondi et al., 2022), being the quadrant most flexible and sensitive to change (Brondi et al., 2022). The graphic representation of the Vergès' Quadrant is detailed in Figure 2.

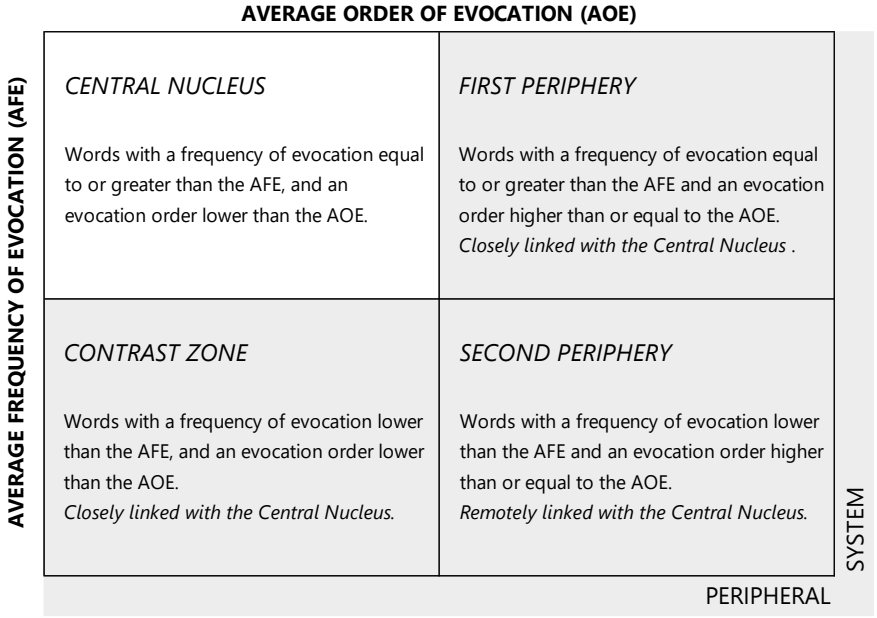


Figure 2. Vergès' four-quadrant technique. Adapted from Joia & Correia (2018).

Similarity Analysis and Content Analysis

The technique of similarity analysis is then used to outline social representations based on graph theory, by analyzing the frequency and the order of evocation via implicative statistics (Flament, 1987; Marchand & Ratinaud, 2012). It is usually used as a checking process for the Vergés' quadrant, and it can be performed through matrix analysis conducted in the open-source software Iramuteq. The similarity analysis helps researchers visually represent the structure of elements, measuring the associations and the level of connection between them (Marchand & Ratinaud, 2012). It allows researchers to determine the proximity between the elements, by identifying the most critical words that could destabilize the entire social representation if removed (Flament, 1981; Vergara, 2005). This approach examines the closeness and connections among a group of elements in the form of maximum trees. Since the number of links between two items increases exponentially with the number of vertices, similarity analysis aims to minimize the number of links to create a connected graph without any cycles (Marchand & Ratinaud, 2012).

Another screening that contributes to the checking process is the content analysis, which systematically examines a communication material, thereby enabling the identification of its underlying

message (Mayring, 2004; Vergara, 2005). The open responses detailing the participants' reasons for choosing the words evoked on the Pix were used as a textual corpus to construct a word cloud. The word cloud technique is a valuable way of summarizing text visually by displaying the most frequently occurring words in a text in a larger font size. That simple quantitative overview is achieved by positively correlating the font size of the depicted tags with the word frequency. Overall, word clouds help to quickly identify the kind of topics present in a text (Heimerl et al., 2014, January).

4. Results

4.1 Sample

Data collection occurred between March and April 2024 through the online platform Qualtrics. The sample of respondents was obtained via professional contacts, with the required profile of having worked, owned, or been a consultant in Brazilian retail companies in the last few years. Direct messages were sent mainly via LinkedIn and WhatsApp groups and single contacts, totaling 500 invitations. Two academic specialists in SRT and word evocation techniques and two specialists in the retail industry validated the questionnaire. A total of 105 final responses were recorded with usable data. A summary of the sociodemographic characteristics of the respondents is presented in Table 5.

Table 5. Sociodemographic Characteristics of the Sample

Sociodemographic Characteristics of the Sample	
Total Respondents	105
Retail Segment	Supermarkets (25%), Wholesalers (13%), C-Stores (11%), Drugstores (9%), Apparel (5%)
Company Size	Large 500+ (64%), Small/Medium 10-99 (17%), Micro (13%), Large 100-499 (6%)
Market Range	National (49%), Regional/State (31%), Local (15%)
Company Headquarter	São Paulo (45%), Rio de Janeiro (28%), Minas Gerais (7%), Rio Grande do Sul (4%)
Job Title	CEO/Director (34%), Manager (26%), Consultant (18%), Coordinator (17%)
Years of Job Experience	20 years or more (69%), 11 to 19 years (21%), 4 to 10 years (6%), 1 to 3 years (3%)
Gender	Male (73%), Female (27%)

4.2 Central Nucleus and Peripheral System in the Vergès' Quadrant

From the 105 questionnaires considered, 525 evoked terms were obtained. The tabulation process conducted in Excel software included the following process: correcting typo errors and the spelling of words, organizing their semantic linkages, considering one version for each word, categorizing them, and translating into English since the questionnaire was conducted in Portuguese. After the words' categorization, 102 different terms were found. Then, one verified which words/ expressions would be placed in the four quadrants of Vergès.

The prototypical analysis of frequency and order of evocation was performed in Excel. For this, the minimum value of the evocation frequency was calculated, considering the median value of all frequencies, which was 11. Thus, to appear in Vergès' quadrants, a word must have been evoked at least 11 times. As no evocation had a total frequency of 11 or 12, the cut-off point was 13, the frequency immediately higher, representing 64.6% of all occurrences. Table 5 details the cut-off calculation.

Table 5. Distribution of the Frequencies of Evoked Words

Number of words	Frequency	Words x Frequency	Accum. Evocations	%
1	104	104	104	19.8%
1	66	66	170	32.4%
1	62	62	232	44.2%
1	40	40	272	51.8%
1	20	20	292	55.6%
1	19	19	311	59.2%
1	15	15	326	62.1%
1	13	13	339	64.6%
1	8	8	347	66.1%
4	7	28	375	71.4%
2	6	12	387	73.7%
2	5	10	397	75.6%
6	4	24	421	80.2%
6	3	18	439	83.6%
13	2	26	465	88.6%
60	1	60	525	100.0%

A final step for assembling the quadrants is the calculation of the Average Order of Evocation (AOE), which represents the weighted average of the average order of evocation of each category evoked, as seen in the subsequent formula:

$$AOE = \frac{(f\ 1st.words \times 1) + (f\ 2nd.words\ \times 2) + (f\ 3rd.words \times 3) + (f\ 4th.words \times 4) + (f\ 5th.words \times 5)}{\sum f}$$

Table 6 lists the frequencies ranked from the highest to lowest and calculations of the AOE for each category considered above the cut-off point. This calculation makes it possible to identify the distribution of the words within the quadrants.

Table 6. Calculation of the AOE of the Vergès' Quadrant's Evocations

	1st Ev. *f1*	2nd Ev. *f2*	3rd Ev. *f3*	4th Ev. *f4*	5th Ev. *f5*	Frequency *∑f*	AOE
1 Speed	41	28	15	12	8	104	2.21
2 Convenience	26	20	7	8	5	66	2.18
3 Ease of use	19	20	9	7	7	62	2.40
4 Security	3	6	13	10	8	40	3.35
5 Innovation	1	4	9	4	2	20	3.10
6 Tax free	2	1	9	6	1	19	3.16
7 Technology	1	4	5	4	1	15	3.00
8 Trust		2	1	7	3	13	3.85

Figure 2 shows the graphic representation of the AOEs, where the lower numbers relate to the first memories and the higher numbers relate to the last recalls.

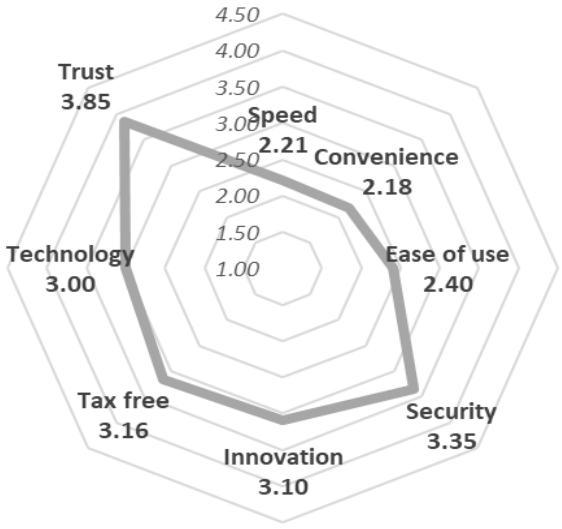


Figure 2. AOE Ranking of the most evoked words.

From the 525 words evoked, eight reached the minimum average frequency of evocation (AFE) of at least 13. These words represent 339 (64.6%) of the total evocations. However, the median of these top eight words was also calculated to determine the cut point of the frequency in the quadrants. The median result was 20, representing the minimum value for a word to be part of the upper quadrants in the Vergès' quadrant. The words are displayed according to their results, which are detailed in brackets in Figure 3.

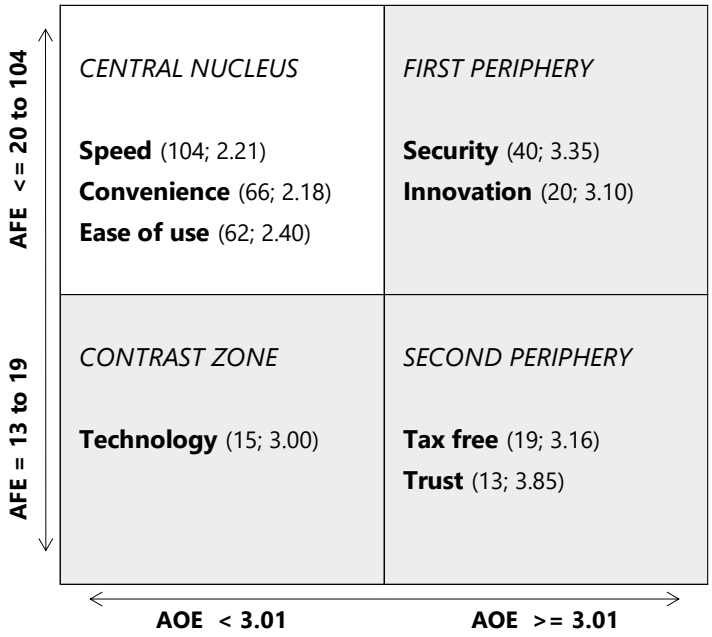


Figure 3. Vergès' Quadrant results accrued from the words evocation technique

4.3 Similarity Analysis and Content Analysis

A social representation is not just a collection of individual elements but rather the interconnection and integration of these elements together. Therefore, identifying a central core

implies extracting its central elements and identifying the connections among them (Abric, 2001; Lin et al., 2013). The analysis of similarity, a quantitative approach that measures the co-occurrence links between segments (Flament, 1981), helped to check the ties among the words evoked, as presented below in the hierarchical tree in Figure 4.

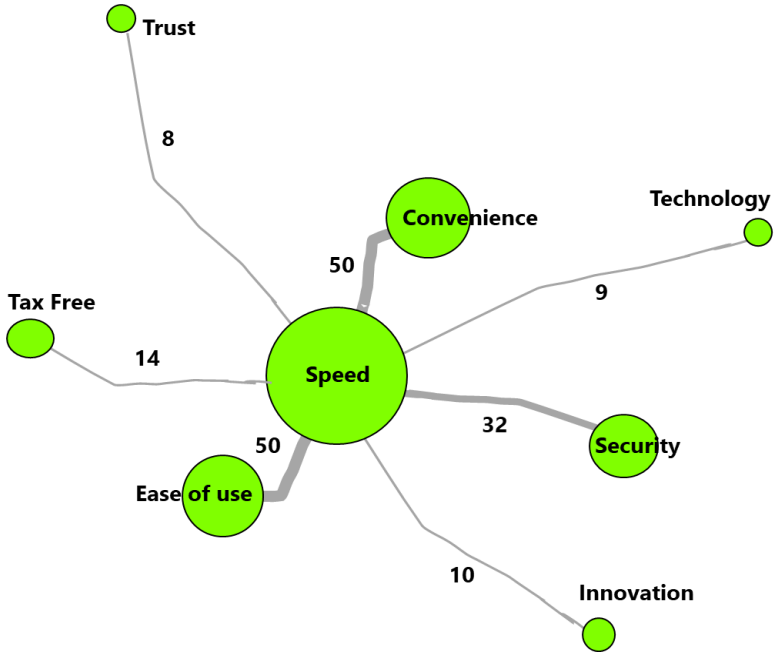


Figure 4. Similarity Tree from Free Evocation Technique

The most relevant elements in the similarity tree present high frequency (circle size) combined with solid co-occurrences (edges/lines) and are precisely the same elements that configure the central nucleus accrued from the Vergès' quadrant– namely, speed, convenience, and ease of use. Accordingly, speed is confirmed as the most meaningful component of the central nucleus, the most potent link representing Pix. Convenience and ease of use present the same number of 50 co-occurrences, although convenience has a broader frequency circle than ease of use, following the same logic of frequency and order of evocation from the Vergès' quadrant.

Then, security appears in the tree with 32 co-occurrences, being placed in the first periphery in the quadrant because of its higher AOE. Innovation was also placed in the first periphery, although, according to the similarity tree, it had a much weaker connection with the main elements. Technology and trust had similar frequencies and connections. However, they were located in separate quadrants as technology had a higher frequency of evocation and a lower AOE, being placed thus in the contrast zone. On the other hand, trust had a higher AOE, being thus placed in the second periphery (peripheral system).

The subsequent analysis considered the textual corpus obtained from the respondents' open answers related to the reasons why they evoked the corresponding terms - which was used to build a words' cloud through the website wordclouds.com. This technique provides a straightforward visualization by displaying the words in different font sizes proportionally to their simple frequencies. The word cloud is depicted in Figure 5.



Figure 5. Words cloud

This assessment analyzed 95 short phrases totaling around 1,000 words after being translated from Portuguese to English. In this analysis, the words were considered as mentioned in the sentences and were not organized into categories as in the previous analysis of the free evocation technique. In the previous screenings, terms such as ease and simple were categorized as ease of use, while speed, agility, and fast were categorized as speed, being convenience alone the word most cited. The sizes associated with these words in the cloud confirm, thus, their placement in the central core. From the remaining words that compose the Vergès' quadrant, security also appears significantly, but innovation, technology and trust do not emerge in prominent positions.

According to the combined analysis of Vergès' quadrant, the similarity tree and the word cloud, innovation is the only element that raises a change in its position in the social representation. Innovation was initially allocated in the first periphery. However, its total frequency is much lower than the other elements in the quadrant and its AOE is within the limit of the quadrants. Moreover, considering the similarity tree and words cloud, innovation does not appear as an essential feature, being thus moved to the second periphery. Figure 6 shows the final social representation of Pix.

CENTRAL NUCLEUS	*FIRST PERIPHERY*
Speed **Convenience** **Ease of use**	**Security**
CONTRAST ZONE	*SECOND PERIPHERY*
Technology	**Innovation** **Tax Free** **Trust**

Figure 6. Social Representation of Pix in the Retail Industry

5. Discussion and Conclusions

The Social Representation of Pix in the Retail Industry

Social representations are the result of a social group's collective memories and experiences, thereby helping name and classify ideas and practices related to various topics in everyday life. As seen from the final Vergès' quadrant in Figure 6, the three ensuing categories compose the central nucleus of the social representation of the Pix, namely speed, convenience, and ease of use. All of them relate to functionalities provided by the system in terms of use and represent the stability of the concept. Even in the vision of retailers' representatives, their central idea on Pix is deeply connected to the customer's perspective and not just the organizational perspective.

Moreover, digital payment and mobile payment have been discussed through the lenses of the intention to adopt technology, usually considering the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Davis, 1989; Venkatesh et al., 2003). TAM model utilizes intentions and behavior as a function of perceived usefulness and ease of use to predict and explain user acceptance of information technologies (Davis & Venkatesh, 1996). Regarding UTAUT, individual's intention to use a particular technology is influenced by performance expectancy, effort expectancy, social influence, and facilitating conditions (Venkatesh et al., 2003; Alkhowaiter, 2020). Indeed, enablers of technology adoption include perceived usefulness, ease of use, trust, security, performance, hedonic motivation, and perceived service, while inhibitors to adopt same include cost of use, perceived risk, complexity, resistance to change, and privacy concerns (Kim et al., 2010; Kajol et al., 2022). Some of these categories are listed as crucial in the conceptualization of Pix. Thus, the meaning of each category composing the Central Nucleus is detailed below. Moreover, the excerpts obtained from the content analysis are also used to help explaining the results.

Speed – Despite their crucial role in digital and mobile payment systems, this construct has received relatively little attention in academic research (Teo et al., 2015). Nevertheless, Yang (2009) believes that the increasing in the speed of transactions would encourage mobile banking advances. Speed is, therefore, pointed out as one of the essential requirements in the payment sector and a critical benefit of m-payment (Chen, 2008; Teo et al., 2015). The official Pix website highlights speed as one of the system's main characteristics: 'Fast- Transactions completed in a few seconds, resources available to the recipient in real time' (BCB, 2024). Speed also appears in the results through synonymous expressions such as fast, velocity, and agility. Speed is often associated to the idea of the convenience of the system or app, highlighting the practicality of the attribute. Moreover, retailers explained that via Pix their business operations are more agile, also pointing out transaction speed as a significant benefit of the tool, as seen in the collected excerpts presented below:

- *Because it is fast and very practical.*
- *Pix's most significant benefit is that it is instant and available anytime.*
- *It has made operations and receipts more agile.*
- *It has made it much faster and more practical to transfer money.*
- *Speed in resolving a payment, whether routine or sporadic.*

Convenience – Convenience is derived from the Theory of Convenience (Teo, 2015), and although it is usually associated with expressions like easy, easiness and simple, the term is more related to use practicality (Jamil & Khan, 2016). Mobile payment users prioritize convenience above everything else because it offers a faster and simpler method for making payments (Kalinic et al., 2019; Kleivene, 2018). Chen (2008) supported the idea that the convenience of having a single payment device to replace multiple payment alternatives has contributed to the benefits of m-payment. Moreover, many studies supported the positive relationship between perceived transaction convenience and innovation

adoption (Teo et al., 2015). Also, retailers recognize convenience as a paramount attribute from their managerial view, when signaling the convenience of payment via QR code or because of the easy way of receiving payments. Indeed, QR Code payment is a prominent feature of Pix, as from a customer-centric perspective, it allows eliminating cash or carrying cards. Pix also brought to the broader public the convenience of a 24/7 payment service (BCB, 2024). All of that can be perceived in the collected excerpts presented below:

- *The convenience of making payments and receiving payments in a simple, practical, and democratic way.*
- *Due to the convenience of immediate payment without needing cash or carrying a card.*
- *Payments at the checkout with a QR code are practical and secure, both for the retailer and the customer.*
- *Essential... because it is practical.*
- *It is practical for the customer. For retailers, it eliminates card operator fees.*

Ease of use – Perceived ease of use refers to the degree to which an individual believes that using a specific system will be effortless (Davis, 1989). In fact, according to TAM, functionality and ease of use are the primary theoretical support for technology diffusion (Davis, 1989). Ease of use is largely cited in the payment literature as several studies apply the TAM/UTAUT approach (Kim et al., 2010; Hanafizadeh et al., 2014; Singh & Sinha, 2020; Ha et al., 2023). Indeed, ease of use is a significant antecedent for mobile payments and positively influences individual's intention to use m-payment services (Chen, 2008; Johnson et al., 2018). This relevant feature is in line with the overall perception of the retailers. Besides, according to the literature, reachability is an essential predictor of ease of use, while ease of use positively influences satisfaction (Alkhowaiter, 2020; Kim et al., 2010). The reachability concept is also related to the Perceived Critical Mass (PCM) theory, which suggests that a small group of initial adopters of a new idea, technology, or innovation can persuade others to join in, leading to rapid growth over time (Oliver et al., 1985). Moreover, PCM is a significant antecedent to users' satisfaction with Pix (Joia et al., 2024). However, satisfaction and reachability were not underscored in the survey. Overall, the results demonstrate the importance of simplifying payments and money transfers to promote positive experiences and ease customer interactions. Furthermore, retailers consider the ease of controlling money flow relevant from the government's perspective. Furthermore, convenience and trust also appear related to ease of use. Gefen et al. (2003) observed that when a system is easy to use, it is classified as convenient, which increases confidence in same. As can be seen, the constructs were related to each other in the excerpts presented below.

- *Due to its ease of operation.*
- *The simplicity of making a payment via Pix.*
- *Easy, practical, and secure.*
- *The large public's access to this agile and easy payment method stimulates sales.*
- *Technology making life easier. Pix is practical and free of charge.*
- *It makes the finance department's job a lot easier.*
- *A government's launch to make it easier to control the flow of money between people and between companies.*

Additionally, in the Peripheral System, five other constructs were associated with the social representation of Pix: security, technology, innovation, tax-free, and trust. The discussion below will group security and trust as well as technology and innovation due to the strong interconnectedness of these constructs in the literature.

Security and Trust – Security significantly influences ICT adoption (Tan et al., 2009). Although digital payment services are more secure than traditional ones, security breaches may have made

consumers wary of using new payment technologies (Johnson et al., 2018). Indeed, more than a decade ago, security was considered a significant barrier to digital adoption and a reason for negatively impacting satisfaction levels among customers (Poon, 2008; Tan et al., 2009). In this sense, providing customer security has been pivotal for mobile payment companies (Luarn & Lin, 2005). More recently, according to the worldwide security systems provider Norton (2024), most banking apps provide several features designed to safeguard customers' sensitive information and identity, surpassing the security measures typically found on traditional banking websites. These cybersecurity features include encryption, frequent software updates, and biometric authentication. However, although many high-end technologies have been developed to enhance security, attacks have become more sophisticated, and cybercrimes have taken the scene, emphasizing the need for robust cybersecurity measures (Chisty et al., 2022; Baddam et al., 2023). One of Pix's challenges was related to security (do Vale et al., 2022). However, regulations and improvements were made to the app, as Pix was the target of many frauds in 2021 (Supervarejo, 2023). In the category related to security, trust is vital for online transactions and a critical factor in determining a well-succeeded strategy for mobile payments (Brown et al., 2003). Banks, cell phone producers and telecommunications operators are the three agents that provide the necessary infrastructure to generate trust in customers (Hanafizadeh et al., 2014). Trust is also related to credibility in the literature, as in the work of Koenig-Lewis et al. (2010) and Hanafizadeh et al. (2014). In the survey, security was much more frequently remembered than trust, although both were in the quadrants with the lowest order of evocation. However, the association between trust and credibility was not perceived in the survey. Retailers also associate the importance of security for both sides, stores and customers. In the case of the stores, for those that still use the technique of 'cash bleeding'[2], Pix solution brings more security due to the decrease in cash, as explained in the excerpts below.

- *It is entirely safe to use this payment/receiving method.*
- *The security of not transacting physical currency.*
- *The first thing that matters in an electronic transaction is security, especially in a financial transaction.*
- *Important for the security of customers and the store (no need to bleed cash).*
- *Payments at the checkout with QR codes are practical and secure, both for the retailer and the customer.*

Technology and Innovation – In a scenario of constant changes, the technology and innovation categories are commonly cited in the digital transformation literature (Vial, 2019; Gong & Ribiere, 2021; Verhoef et al., 2021). However, in the case of technology, in the mobile payment literature, the word often appears related to its adoption and use (Hanafizadeh et al., 2014; Alkhowaiter, 2020) as the primary and indispensable tool for digital transformation, while innovation represents the novelties and disruptions such advances have brought to society (Demirkan et al., 2016). In social representation studies, technological aspects often take a sociomaterialist approach, where social practices are closely connected with technologies and infrastructures (Orlikowski, 1996). In this sense, social practices are influenced by technology, and technology also influences them (Orlikowski & Scott, 2008). This means that technology alone is insufficient to drive the entire transformational process (Nadkarni & Prügl, 2021). In the survey, the aforementioned categories appear in the peripheral system, given their smaller frequencies, representing thus possible mutable features in the context of the social representation. However, technology had a better result regarding the order of evocation than innovation, which placed them into different quadrants. The responses about the choice of evoking technology and innovation were few, as they were less frequent categories in the quadrant, as put forth below:

[2] Cash bleeding refers to the process of withdrawing cash from the point of sale (POS) when it reaches a certain amount or when the operator's work shift changes, whereby the amount must be counted and stored in a safe place (Infovarejo, 2021).

- *Using technology to make life easier.*
- *Because of the modernity and ease of use of this technology.*
- *Perception of the innovation generated by Pix.*

Tax Free – This category represents the importance of total absence of taxes or charges in digital financial services enabled by Pix. According to Hanafizadeh et al. (2014), some digital tools or apps are charged, and this cost of use is a common barrier to adopting new technologies. However, the topic is not mentioned in the same intensity as the other categories in the literature on digital payments. In this research, retailers acknowledge the characteristic of free-of-charge status as paramount for customers, especially those from lower incomes. Moreover, Lobo & Brandt (2021) endorse Pix's role in facilitating payment by permanently being free for users, positively impacting financial inclusion. Although tax free appears in a quadrant that demonstrates the mutable characteristics of the social representation and not in the central core, retailers answered that the gratuity of the service is a critical feature for the use and success of Pix in the current scenario. According to the Brazilian Society of Retail and Consumption (SBVC), more and more Brazilians are getting used to Pix as a form of payment, mainly because it reduces costs for both the user and the establishment offering it. Additionally, establishments have lower costs when been paid via Pix than via traditional card payment (Supervarejo, 2023). Because of this, SBVC is confident that Pix payments will continue to grow in terms of user base and acceptance within the retail sector. The following quotes describe the importance ascribed to the gratuity of Pix and consider the importance of same for low-income customers:

- *Because it was launched in favor of the population, without charging fees.*
- *The fact that there are no fees is crucial for lower-income customers, many of whom do not have a card.*
- *There are no fees, and everyone accepts them.*
- *I believe it is a more cost-effective process because no fees are involved. Especially because nowadays, we have taxes on everything when it comes to e-commerce.*

Ultimately, a category that calls attention to its absence in the social representation of Pix is privacy. Although security is an essential aspect that composes retailers' main memory, privacy concerns were not mentioned in the survey although this category is a recurrent theme in the literature on mobile payments (Luarn & Lin, 2005; Nasri & Charfeddine, 2012; Kajol et al., 2022; Baddam et al., 2023). More recently, this has become a subject associated with discussions about the use of technologies such as Internet of Things, digitization of personal data, economics of IT and Artificial Intelligence (Gozman et al., 2020; Leidner & Tona, 2021; Brynjolfsson et al., 2021; Zhdanov et al., 2022). Privacy has often been merged with security, being a crucial concept that supports freedom, justice, and democracy (Wahlstrom et al., 2020). Nevertheless, the lack of privacy in data transactions is a significant concern, although the subject is still very topical in the literature, especially given the increasing use of personal information for technological purposes. Indeed, Brynjolfsson et al. (2021) posited the question of which principles should be determined considering data ownership and privacy.

5.1 Research Limitations, Contributions, and Future Agenda

As with any academic research, this study also has some limitations. Considering the theory used, there are critics of social representations who allege a lack of a critical agenda. However, many criticisms stem from the challenge of comprehending the complex and dynamic relationship between individual agency and social structure, which is central to the SRT (Voelklein & Howarth, 2005).

Moreover, interpreting qualitative information and classifying the evoked words is a process that involves a subjective approach and manual work from the researcher in a way that may have some bias

- being, therefore, a limitation in the study. The research also considers the interpretation of retailers from different parts of Brazil and diverse sizes of companies. However, it may not have covered perceptions from other specific retail segments, such as sports goods or luxury—thus, this can be further researched to deepen knowledge in specific retail segments.

Regarding the study's contribution, the first one is to expand the knowledge and applicability of perceptions and practices related to the field of organizational digital transformation, especially considering digital payment. It contributes, thus, to the conceptualization of digital transformation, a multifaceted construct (Vial, 2019). This study also serves as a guide for practitioners to evaluate digital transformation efforts related to digital and mobile payments in their companies. Pix seems to have fulfilled the role of spreading digital technologies and consequently has been part of digital transformation initiatives among retailing companies.

Moreover, Pix has been recognized as an essential player for financial inclusion (Lobo & Brandt, 2021). This way, the third contribution of this study is to deepen the understanding of digital and mobile payments in an emerging country's context from retailing companies' perspective. As a last contribution, the investigation expands the use of SRT in academia, bringing new insights on its use in ICT and IS studies. Besides, future steps in that realm include further investigation into how organizations have responded to the challenge of digital transformation, considering the adoption of digital and mobile payments. In fact, this study did not consider the impacts of COVID-19 on the evolution of digital payments and other financial services in emerging economies – thus, this configures a research topic to be addressed.

Future research may also amplify the debate about financial inclusion in developing countries. Expanding knowledge about the distance between financial inclusion and the promotion of true social inclusion is also a topic that needs to be explored further. Furthermore, future studies may address the perspective of low-income individuals on Pix, as well as their relationship with this form of payment, deepening the understanding of the main problems and challenges that this population has faced when dealing with the digital transformation of media. of payment. Lastly, financial institutions and retail organizations play a critical role in the transformative power of digital technologies in this sector and should actively engage in promoting financial and digital literacy among citizens, especially in emergent economies. This work aims to have been a bridge between academia, financial institutions, and retail organizations in Brazil.

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8. Final Considerations

The shared understanding of DT as a phenomenon and the unveiling of the theoretical foundations encompassing the multiple perspectives associated with same (including social structures and technology) are still exploratory fields in academia (Gudergan & Mugge, 2017). Since DT is a continuous process, this dissertation aims to understand how organizations navigate the DT journey, from preparation and readiness for change through defining critical success factors to understanding how a DT initiative is represented and assimilated in a business environment.

In this sense, the research has brought the answers to the proposed question, *"How Digital Transformation has advanced within organizational environments, from evaluating their attributes for readiness to understanding how they manage and assess critical success factors in actual DT processes?"* and its related objectives were accomplished— reviewing the literature on digital readiness to structuring an integrative framework of organizational DT readiness; analyzing the critical success factors of DT in organizations, ranking them in order of importance; and analyzing how an existing DT process in retail organizations is perceived by exploring the details of a new digital payment method, Pix.

Therefore, the three papers connect under the umbrella theme of DT, proceeding one level down from the primary theme, where DT's unique and evolving phenomenon within organizations is analyzed. The research focused on the retail sector within a developing country, with comprehensive research involving three sequential stages of investigation, with its underlying overall objectives detailed in Figure 2.

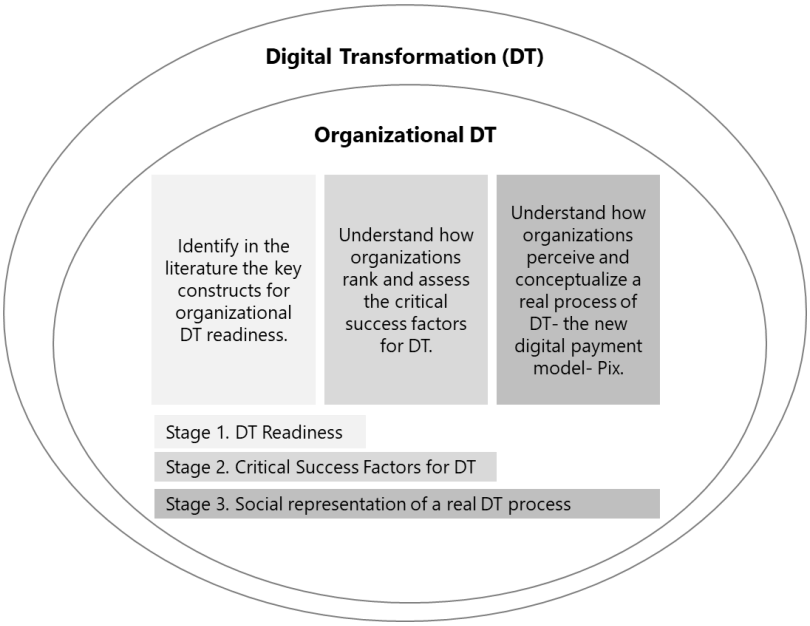


Figure 2. Digital Transformation in Organizations: Dissertation Plan

The research was conducted in Brazil, a country considered an emerging economy embracing two worlds within one - highly developed and very poor societies coexisting. The focus of this work was in the retail sector in order to understand the organizational compositions, conceptualizations, and practices in this industry, as previously justified. Since the opening of the retail sector to foreign direct investment in the 1990s, Brazil and other Latin American countries expected the entry and expansion of retail chains to stimulate the development of a current scenario most determined by family-run companies and independent stores (Reardon et al., 2003; Vries, 2008). Moreover, some companies have started adopting new ICT, and Brazilian retailers have been trying to specialize in searching for the latest innovations and new business processes.

In the first paper, one proposed an integrative framework for assessing the ODTR based on a review and analysis of the extant scientific literature on this subject, revealing an overlap between the definitions of e-readiness and digital readiness constructs. Besides, the work also unveiled the lack of a unified and monosemic definition for "digital transformation readiness," especially in the organizational realm. It is also noteworthy that several relevant concepts are directly involved with ODTR, which is corroborated by the evidence found in the related literature. These concepts are related not only to ICT but also to myriad organizational dimensions, such as resources, processes, organizational culture, competitiveness, management, human competencies, business models, and sustainability, thus confirming the theme's polysemic characteristic.

Besides, the second paper, via the Delphi method, unveiled that the top positions of the critical success factors for organizational DT are related to the Management Capability dimension. This perspective emphasizes that investing solely in ICT is insufficient for the success of DT initiatives, and underscores the crucial role of top management's mindset in driving DT. The second most relevant critical success factor considered the discussions over omnichannel retail and its new business opportunities, a significant trend in Europe and North America (Galipoglu et al., 2018; Sousa et al., 2021). However, Technology, far from the top positions, corroborates with the sociomaterialist theories in IS studies— which refutes the idea of having technology as paramount above other actors (Hassan & Hovorka, 2011) — and shall indicate that the retail business in the context of an emerging economy may have reached a level of DT maturity comparable to that of its more developed counterparts.

Retailers also ascribed low importance to factors related to adopting the most recent innovative technologies, such as AI or augmented reality. The latest technological novelties are far from being implemented on a large scale in the Brazilian retail industry, where artificial intelligence (AI) in the

retail sector has only been incorporated by 16% of companies (Exame, 2023), which highlights the gap between the operation in developed economies and their counterparts in developing economies. This second paper provides a new configuration of factors and their respective dimensions, that are considered paramount for the success of DT initiatives in companies. It also aims to encourage debate not only about what they are, their descriptions, and conceptualization but also to broaden the understanding of how they are being managed or, at least, whether or not they are being implemented in organizations.

The dissertation's third paper investigates the perceptions about how organizations deal with an actual DT initiative, by understanding the retail's social representation of Pix, a peer-to-peer digital payment system. The results demonstrated that the primary categories are related to attributes that are often discussed in the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) models (Davis, 1989; Venkatesh et al., 2003), considering the intention to adopt the technology. Speed, convenience, and ease of use are the three categories that compose the core meaning of the social representation of Pix. The research's retail vision has a central idea that is deeply connected to the customer's perspective, and not only from the organizational perspective. The other main characteristics involved in the digital transformation process of mobile payment considered by retailers in Brazil refer to security and trust, technology and innovation, and tax-free, pointed as the peripheral understanding of the Pix phenomena.

All in all, the trajectory of the three papers under the umbrella theme of DT improves the understanding of perceptions and practices regarding organizational digital transformation. These papers contribute to the understanding of the evolutionary process of ODT by providing three-phased analyses: a new framework for evaluating DT readiness and initiatives in organizations, a ranking of critical factors for assessing and evaluating the success of DT initiatives, and insights into the perceptions and conceptualization of a recent, widespread DT initiative implemented in organizations. Moreover, the work unveils new findings about the multifaceted construct of DT in all papers. It also provides comprehensive discussions for academics and practitioners to evolve the ODT discussions in the context of an emerging economy.

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